



Community Development – Zoning Division

John Pederson – Division Manager

ZONING CASE

Z-5-2026

SITE BACKGROUND

Applicant	The Revive Land Group, LLC
Phone	Contact Zoning Office
Email	Contact Zoning Office
Representative Contact	J. Kevin Moore
Phone	Contact Zoning Office
Email	Contact Zoning Office
Titleholder	Estate of Lucille Garrett a/k/a Leathy Lucille Garrett a/k/a Leathy Lucille Price Garrett and William D. Brown
Property Location	Located on the east and west sides of Ernest Barrett Parkway and the western terminus of Old Horseshoe Bend Road
Address	None assigned
Access to Property	Ernest Barrett Parkway and Old Horseshoe Bend Road

QUICK FACTS

Commission District	4 - Sheffield
Current Zoning	R-20
Current Use of Property	Vacant
Proposed Zoning	RA-5
Proposed Use	Single-family community
Future Land Use	MDR
Site Acreage	10.17
District	19
Land Lot	691
Parcel #	19069100010, 19069100070
Taxes Paid	Yes

FINAL ZONING STAFF RECOMMENDATIONS

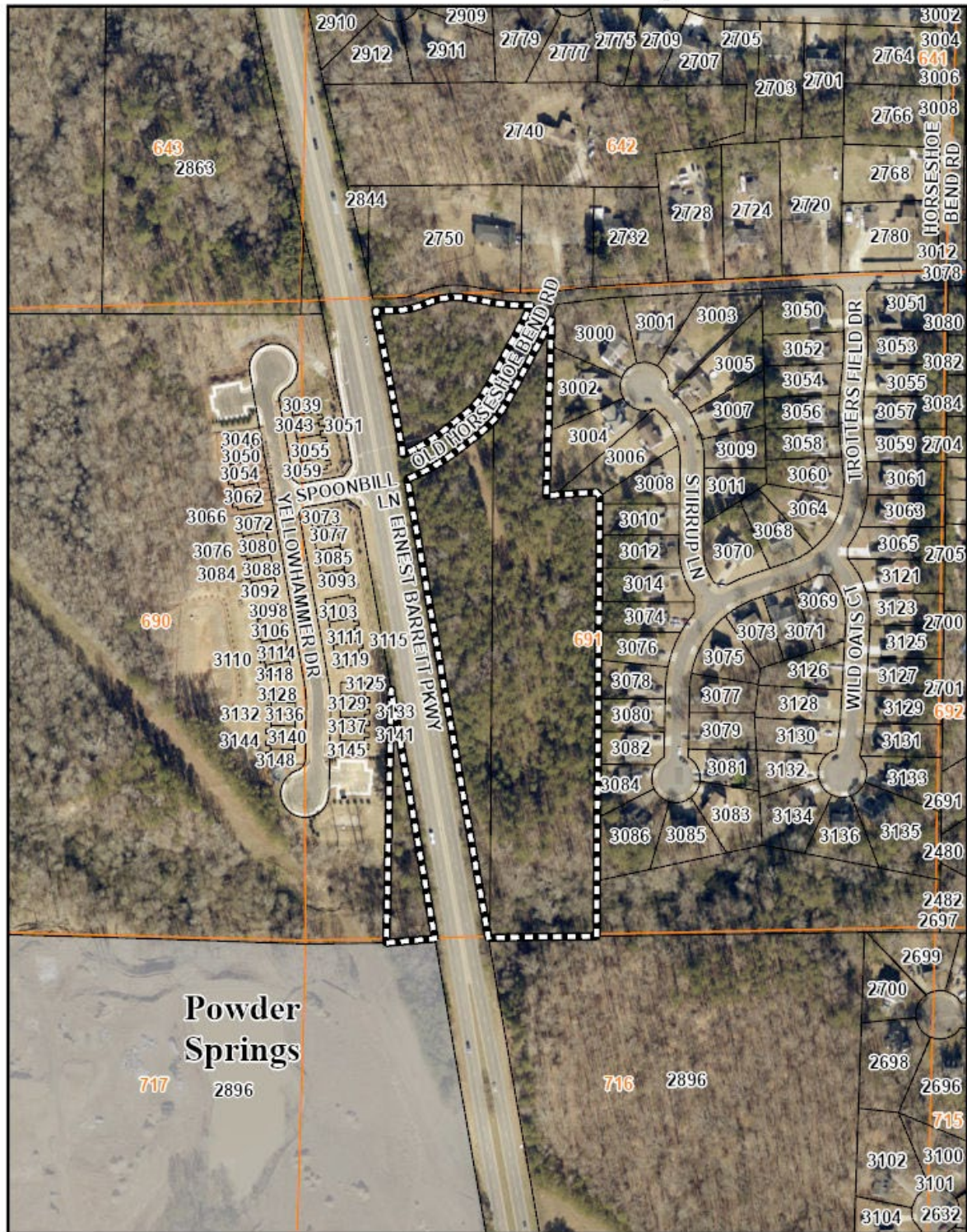
(Zoning staff member: Cameron Jones, CNU-A)

Based on the analysis of this case, Staff recommends **APPROVAL** subject to the following:

Z-5-2026 DEPARTMENT COMMENTS – Zoning Division

1. District Commissioner to approve final site plan;
2. Maximum density of 5 units per acre;
3. Variances stated in the Zoning comments;
4. Fire Department comments and recommendations;
5. Stormwater Management Division comments and recommendations;
6. Water and Sewer Division comments and recommendations; and
7. Department of Transportation comments and recommendations.

Z-5 2026 Aerial Map



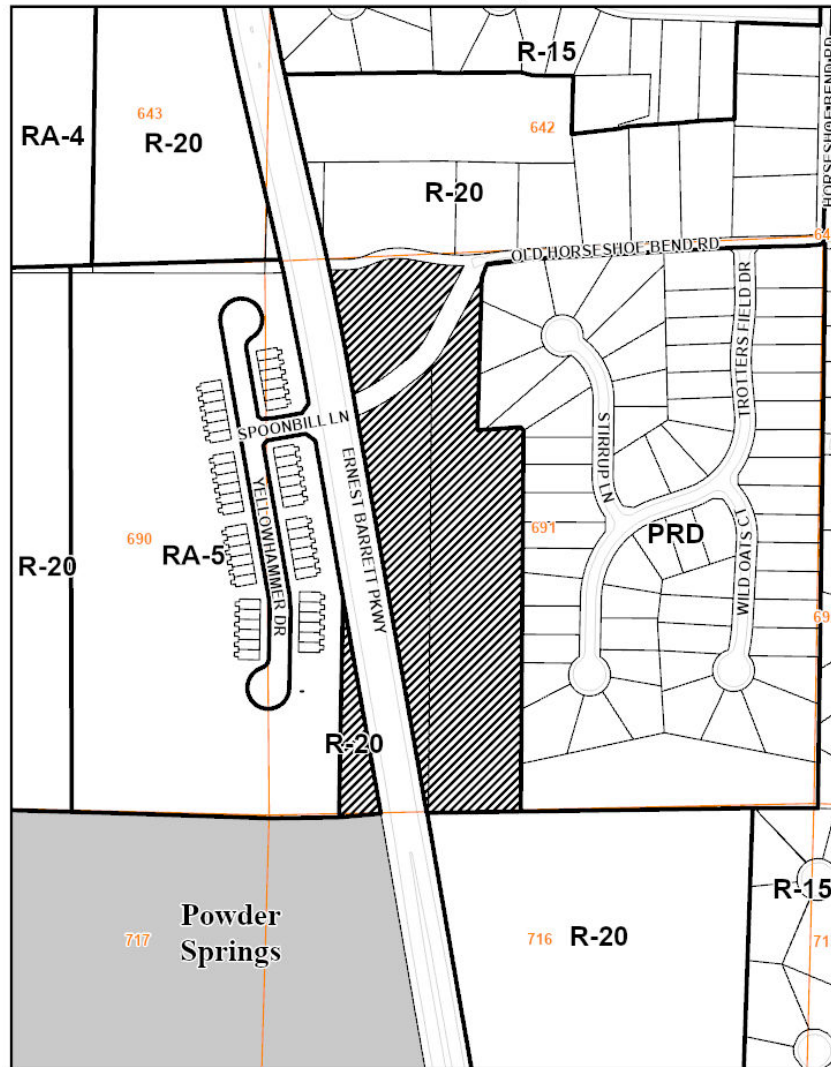
Z-5-2026 DEPARTMENT COMMENTS – Zoning Division

North

Zoning: R-20 / Single-Family Homes

Future Land Use: LDR

Z-5 2026 Map



WEST

Zoning: RA-5 /
Attached
Townhomes

**Future Land
Use:** MDR

EAST

Zoning: PRD /
Single-Family
Homes

**Future Land
Use:** MDR

SOUTH

Zoning: City of Powder Springs, R-20 / Vacant Lot

Future Land Use: MDR

Z-5-2026 DEPARTMENT COMMENTS – Zoning Division

Current zoning district for the property

The R-20 district is established to provide locations for single-family residential uses or residentially compatible institutional and recreational uses which are within or on the edge of properties delineated for any residential category as defined and shown on the Cobb County Comprehensive Plan: A Policy Guide, adopted November 27, 1990. When residentially compatible institutional and recreational uses are developed within the R-20 district, they should be designed and built to ensure intensity and density compatibility with adjacent single-family detached dwellings and otherwise to implement the stated purpose and intent of this chapter.

Requested zoning district for the property

The RA-5 district is established to provide locations for the development of affordable single-family detached or attached residential dwelling units, including the combination of duplexes, triplexes and quadraplexes. The dwelling units are to be designed so as to be placed on an individual lot attached to another dwelling unit or on an adjoining lot where the units will be attached by a common party wall. This residential use is designed to be located within or on the edge of properties delineated for any residential categories as defined and shown on the Cobb County Comprehensive Plan: A Policy Guide, adopted November 27, 1990. When residentially compatible institutional and recreational uses are developed within the RA-5 district, they should be designed and built to ensure intensity and density compatibility with adjacent single-family detached dwellings and otherwise to implement the stated purpose and intent of this chapter.

Summary of the applicant's proposal

The applicant is requesting to rezone to the RA-5, single-family attached/detached residential district, to develop a 43-unit single-family community on a 10.17-acre site. The homes will be a minimum of 1,600 square feet and proposed to be traditional/craftsman in architecture.

Residential criteria

Allowable units as zoned: 16

Proposed # of units: 43

Net density: 4.44

Increase of units: 27

Acres of floodplain/wetlands: 0.956

Impervious surface shown: Maximum 40%

Z-5-2026 DEPARTMENT COMMENTS – Zoning Division

Are there any zoning variances?

Yes;

1. Reduce the minimum lot size for all lots from required 7,000 square feet to 3,700 square feet;
2. Reduce rear yard setback on an arterial road from required 40 feet to 30 feet;
3. Reduce the rear setback for all interior lots from required 30 feet to 20 feet;
4. Reduce the front setback for all interior lots from required 20 feet to 15 feet;
5. Reduce the major side setbacks for all interior lots from required 20 feet to 15 feet;
6. Reduce the minimum lot width at front setback and the minimum public road frontage for all interior lots from required 70 and 50 feet to 41 feet;
7. Reduce the minimum width between buildings from required 15 feet to 10 feet; and
8. Increase the maximum allowable impervious coverage from required 40% to 45%.

12/17/25

The proposed zoning case must address the comments listed below by prescriptive code or equivalency, as approved by the Cobb County Fire Marshal's Office

Approval will be based on Professional Engineer-stamped plans that signify overall plan conformance to all Georgia State and Cobb County Codes. Design Professional and Developer maintain the sole responsibility for the design and for correcting all errors, omissions, problems, and code violations (if any) exposed during construction AFTER authorization by Cobb County.

FIRE DEPARTMENT ACCESS:

Fire apparatus access roads shall extend to within 150 feet of all portions of the facility or any portion of the exterior wall of the first floor (State Modifications IFC 503.1 2018 Edition and Cobb County Development Standards 402.11). A ten-foot separation, open to sky, is required in order to measure between structures. (CCDS 402.11)

All access roads shall meet the American Association of State and Highway Transportation Officials (AASHTO) design manual live load standard HS20 (75,000 lbs.) with an unobstructed width of not less than 20 feet, 35 foot inside radius, 50 foot outside turning radius and unobstructed vertical clearance of not less than 13 feet 6 inches. (Cobb County Development Standards 402.11)

Maximum slope of the access road in reference to the apparatus is 10% Front to Back and 5% Side to Side. Maximum grade of roadways leading to Fire Access roads refer to the Cobb County Development Standard Section 400: 14% for Non-Residential. Maximum angle of departure is 8.5%.

Dead-end access roads more than 150 feet shall be provided with a turn-around (IFC Chapter-5, 120-3-3 Rules and Regulations, CCDS Section-401 & 402). Refer to APPENDIX-D for dimensional criteria for turn-arounds in commercial developments.

FIRE HYDRANT:

The maximum distance of a hydrant to the most remote portion of a structure shall be 500 feet. Distance shall be measured around the structure and from the fire apparatus access road. A 500-foot radius around the hydrant is not an acceptable measurement for distance. Hydrants on the opposite side of a main ROW is not an acceptable measurement of distance. (Cobb County Development Standards 410.5.2) (Cobb County Development Standards 410.5.2)

Where a fire hydrant is located on a fire apparatus access road, the minimum road width shall be 26 feet for a 40-foot length centered on the fire hydrant, exclusive of shoulders (see Figure D103.1 in Appendix D).

(continued)

GUEST PARKING:

Occupant parking shall be installed as required by zoning and additional guest parking shall be required as follows:

- Where driveways to two car garages exceed 50 feet in length, no additional guest parking is required.
- Where driveways to two car garages are at least 22 feet long and 20 feet wide, additional parking shall be required at .5 spaces per dwelling unit.
- Where driveways to two car garages are less than 22 feet long and 20 feet wide, additional parking shall be required at 1 space per dwelling unit.
- Where only single car garages are provided, additional parking shall be required at 2 spaces per dwelling unit.
- Guest parking spaces must be evenly distributed throughout the project. (Cobb County Development Standards 401.8.2)

SITE PLAN APPROVAL:

Plans must be submitted to the Cobb County Fire Marshal's Office for fire site plan approval.

Jeff Byrd

(678) 229-7059

Jeffrey.Byrd@cobbcounty.gov

Z-5-2026 DEPARTMENT COMMENTS – Cemetery Preservation

No comment

2/3/2026

Approval of this petition will cause concern for CCSD, as it will result in an impact on the enrollment of one (1) school already over capacity.

School	Student Capacity	School Enrollment	Capacity Status
Hollydale ES	862	538	324 under capacity
Smitha MS	1137	806	331 under capacity
Osborne HS	2412	2707	295 over capacity

12/26/2025

Comprehensive Plan Future Land Use Designation: Medium Density Residential

The subject tract is within the Medium Density Residential (MDR) future land use category. The purpose of MDR is to provide for areas that are suitable for moderate density housing between two and one-half (2.5) and five (5) dwelling units per acre. This category presents a range of densities.

Future Land Use Sub-Category? N/A☒ Consistent ☐ Inconsistent

House Bill 489 Intergovernmental Agreement Zoning Amendment Notification

Is the proposal within one-half mile of a city boundary? ☒ Yes ☐ No
If yes, which City: Powder Springs
Was the city notified? Yes

Specific Area Policy Guidelines: ☐ Yes ☒ No
Policy:

Masterplan/ Corridor Study: ☐ Yes ☒ No
Name of Masterplan/Corridor Study: N/A

Design guidelines area? ☐ Yes ☒ No
If yes, design guidelines area? N/A
Does the proposal plan comply with the design requirements? N/A

Is the property within an Opportunity Zone? ☐ Yes ☒ No
If yes, Opportunity Zone? N/A

(The Opportunity Zone is an incentive that provides \$3,500 tax credit per job in eligible areas if two or more jobs are being created. This incentive is for new or existing businesses)

Is the property within an Enterprise Zone? ☐ Yes ☒ No
If yes, Enterprise Zone? N/A

(The Enterprise Zone is an incentive that provides tax abatements and other economic incentives for qualifying businesses locating or expanding within designated areas for new jobs and capital investment)

Is the property eligible for the Façade Improvement Program? ☐ Yes ☒ No

(The Façade Improvement Program is an incentive for owners and tenants to enhance the appearance of buildings. The CDBG program provides the funding. Properties must be either on the current inventory of redevelopment sites or in a corridor study area and be in a census tract with at least 51% low and moderate income. The program serves to improve the economic viability of these areas.)

(continued)

Is the property eligible for incentives through the Commercial and Industrial Property Rehabilitation Program?☐ Yes ☒ No

(The Commercial and Industrial Property Rehabilitation Program is an incentive that provides a reduction in ad valorem property taxes for qualifying redevelopment in eligible areas)

Note: For more information on incentives, please call the Community Development Agency- Economic Development Division at 770-528-2018 or find information online at www.cobbcounty.org/econdev.

Special District:

Is this property within the Cumberland Special District #1 (hotel/motel fee)?

☐ Yes ☒ No

Is this property within the Cumberland Special District #2 (ad valorem tax)?

☐ Yes ☒ No

Dobbins Air Reserve Base Zones:

Is the property within the Dobbins Airfield Safety Zone?

☐ Yes ☒ No

Is the property within the Clear Zone (CZ)?

☐ Yes ☒ No

Is the property within 3000' of Dobbins ARB?

☐ Yes ☒ No

Is the property within the Accident Potential Zone (APZ I)?

☐ Yes ☒ No

Is the property within the Accident Potential Zone II (APZ II)?

☐ Yes ☒ No

Is the property within the Noise Zone?

☐ Yes ☒ No

Is the property within the Bird/Wildlife Air Strike Hazard Area (BASH)?

☐ Yes ☒ No

Historic Preservation:

After consulting various county historic resources surveys, historic maps, archaeology surveys and Civil War trench location maps, staff finds that no known significant historic resources appear to be affected by this application.

Z-5-2026 DEPARTMENT COMMENTS – Water and Sewer

Water comments:

At development: ☒ YES ☐ NO

Fire flow test required: ☒ YES ☐ NO

Size/location of existing water main(s): 12" on Ernest Barrett Pkwy and 8" on Old Horseshoe Bend Rd

Additional water comments: Dual water feed will be required

Sewer comments:

In the drainage basin: ☒ YES ☐ NO

At development: ☒ YES ☐ NO*

Approx. distance to nearest sewer: On-site

Estimated waste generation (in G.P.D.): Average daily = 9,000

Peak flow = 22,500

Treatment plant: South Cobb WRF

Plant capacity: ☒ Yes ☐ NO

Projected plant availability: ☐ 0-5 years ☐ 5-10 years ☒ over 10 years

Off-site easement required: ☐ YES* ☒ NO

Line capacity study required: ☐ YES ☒ NO

Letter of allocation issued: ☐ YES ☒ NO

Septic tank recommended by this department: ☐ YES ☒ NO

Subject to Health Department approval: ☐ YES ☒ NO

Additional sewer comments:

*If off-site easements are required, the developer/owner must submit easements to CCWS for review and approval as to form and stipulations prior to the execution of easements by the property owners. All easement acquisitions are the responsibility of the developer/owner.

Note: These comments only reflect what facilities were in existence at the time of this review. Developer may be required to install/upgrade water mains based on fire flow test results or Fire Department code. This will be addressed in the Plan Review process. The developer/owner will be responsible for connecting to the existing county water and sewer systems, installing and/or upgrading all outfalls & water mains, obtaining onsite and/or offsite easements, and dedication of onsite and/or offsite water and sewer to Cobb County as may be required. Rezoning does not guarantee water/sewer availability or capacity unless so stated in writing by the Cobb County Water System.

Roadway	Roadway classification	Speed limit (mph)	Jurisdictional control	Min. R.O.W. requirements
Barrett Parkway	Arterial	45	Cobb County	100'

Roadway	Location	Average daily trips (vpd)	Level of service
Barrett Parkway	North of Horseshoe Bend Road	21,500	D

vpd = vehicles per day

Based on 2024 AADT counting data taken by GDOT, as published on their website for Barrett Parkway.

Planning Level of Service based on available Average Daily Trips using GRTA guideline thresholds. Classification thresholds for LOS A and LOS B are not available for arterial and major collector roads from this data source.

LOS C or D is acceptable based on GDOT Design Policy Manual criteria.

ITE Land Use	Description	Expected Size (DU)	AM			PM			Daily (vpd)
			In (vph)	Out (vph)	Total (vph)	In (vph)	Out (vph)	Total (vph)	
210	Single Family Homes	43	7	23	30	25	15	40	405

vph = vehicles per hour, DU = dwelling unit

Development gross trip generation based on available development size and rates from the Institute of Transportation Engineers (ITE) Trip Generation, 11th edition.

Comments and observations

Barrett Parkway is classified as an arterial roadway and according to the available information the existing right-of-way does meet the minimum requirements for this classification.

In accordance with Cobb County Code 134-121 and based on site plan dated November 20, 2025, this development does not require a traffic study submittal. However, Cobb DOT has requested a traffic study in accordance with Cobb County Code 106-11, because Ernest Barrett Parkway is a limited access highway that required the Board of Commissioners approve for new access.

A Submittal 1 traffic study assumptions memorandum (dated October 14, 2025) was submitted by this developer at this location. Cobb DOT provided comments (dated November 3, 2025) and approval to the applicant team to proceed with the full Submittal 2 traffic study. A traffic Submittal 2 (dated November 17, 2025; received December 4, 2025) was received for this development. Cobb DOT then received an updated Submittal 2 on December 30, 2025. Cobb DOT provided comments for the updated Submittal 2 on January 7, 2026.

The December 30th traffic study is **approved** by Cobb DOT and the traffic study requirement can be considered **satisfied** for this application.

(continued)

Cobb County DOT supports abandonment of the existing right of way as shown in the proposed site plan. This property owner must coordinate with Cobb County DOT regarding the abandonment and subsequent purchase of the Old Horseshoe Bend Road right-of-way. This will need to be addressed between the applicant and the County before completion of the Plan Review process.

Recommendations

THE RECOMMENDATIONS BELOW SHALL BECOME STIPULATED REQUIREMENTS UPON INCLUSION WITH AND APPROVAL BY THE BOARD OF COMMISSIONERS OF THIS APPLICATION:

1. Recommend applicant verify that minimum intersection sight distance is available for the Barrett Parkway access, and if it is not, implement remedial measures, subject to Cobb DOT's approval, to achieve the minimum requirement of 500'.
2. Recommend Barrett Parkway access include a deceleration (right turn) lane to address safety because Barrett Parkway is an arterial roadway. This recommendation is pursuant to Cobb County Development Standard 402.9, Deceleration Lanes/Roadway Improvements Along Existing Streets, and Cobb County Development Standard Detail 401A, Deceleration Lanes. This recommendation is addressed conceptually in the site plan dated November 20, 2025. However, this site plan does not appear to fully comply with design requirements for this recommendation. Recommend location and design be determined during plan review, subject to Cobb DOT approval.
3. If the development is approved by the Board of Commissioners, recommend applicant submit a plat of survey and legal description of proposed right of way abandonment area to Cobb DOT to abandon and dispose of right of way in accordance with OCGA Section 32-7.
4. Recommend payment of fee in lieu for required sidewalk, curb and gutter installation along Barrett Parkway frontage. This recommendation will be finalized in plan review, subject to Cobb County DOT approval.

(continued)

5. Recommend a minimum of 50 feet straight-line distance (from end of curb return to start of adjacent curb return) between any residential driveways and interior roadways, adjacent curves, on-street parking or driveways. This recommendation is pursuant to Cobb County Development Standard 402.7.1, Residential Driveways and Cobb County Development Standard Detail 116R, Residential Driveway Standard. However, because of the tight home spacing, Cobb DOT notes that this standard cannot be met for adjacent internal driveways with the proposed site plan (dated November 20, 2025). The failure to meet this standard for internal driveway spacing is noted because it will preclude the interior roadways from being accepted by the County as public streets at a future date.
6. Recommend curb and gutter along both sides and sidewalk along one side of proposed internal roadway. The site plan (dated November 20, 2025) appears to comply with this recommendation.
7. Recommend extending uninterrupted access (distance to the first internal parking space or drive aisle) to a minimum of 50 feet from the edge of right-of-way line. This recommendation is to allow for vehicle stacking and is pursuant to Cobb County Development Standard 402.8, Non-Residential Driveways and Cobb County Development Standard Detail 116. Recommend final location and design be determined during plan review, subject to Cobb County DOT approval.
8. Recommend mail kiosk for proposed development be located off the right-of-way and either on the inward traffic flow or adjacent to off-street guest parking. This recommendation is to address safety because of the potential of illegal parking movements crossing opposing traffic. The site plan (dated November 20, 2025) appears to comply with this recommendation.
9. All signage and landscaping should be installed off county right-of-way and where it will not impede the line of sight for the driveways or roadways.
10. Recommend 10' no access easement on Barrett Parkway and frontage along Old Horseshoe Bend Road.
11. Recommend applicant be required to meet all Cobb County Development Standards and Ordinances related to project improvements.

Z-5-2026 DEPARTMENT COMMENTS – Zoning Division

STAFF ANALYSIS

Per section 134-122 of the Official Code of Cobb County, below is a written zoning analysis relating to the following (question in bold; the answer is not bold):

A. Whether the zoning proposal will permit a use that is suitable in view of the use and development of adjacent and nearby property;

It is Staff's opinion that the applicant's rezoning proposal will permit a use that is suitable in view of the use and development of adjacent and nearby property if the density is limited to five (5) units per acre. There are single-family residential developments surrounding this property with lower densities to the east and west.

B. Whether the zoning proposal will adversely affect the existing use or usability of adjacent or nearby property;

It is Staff's opinion that the applicant's rezoning proposal will not adversely affect the existing use or usability of adjacent or nearby properties if the density is limited to five (5) units per acre. The proposed development would have a slightly higher impact on surrounding development. The subdivision Trotter's Field, located to the east, is developed at three (3) units per acre and the Barrett Cove Townhome community, located to the west, is developed at three point eight (3.8) units per acre.

C. Whether the zoning proposal will result in a use which will or could cause an excessive burdensome use of existing streets, transportation facilities, utilities, or schools;

It is Staff's opinion that the applicant's rezoning proposal will not result in a use which could cause an excessive or burdensome use of existing streets, transportation facilities, utilities, or schools if the density is limited to five (5) units per acre. This opinion can be supported by the department comments contained in the analysis.

D. Whether the zoning proposal is in conformity with the policies and intent of the comprehensive plan;

It is Staff's opinion that the zoning proposal would be in conformity with the Cobb County Comprehensive Plan if the density is limited to five (5) units per acre. The property lies within the Medium Density Residential (MDR) future land use category. The MDR designation allows for moderate density housing between two and one-half (2.5) and five (5) dwelling units per acre. The project is proposed at 4.4 units per acre which would conform to the Medium Density Residential (MDR) future land use category, but be slightly higher than the surrounding developments.

Z-5-2026 DEPARTMENT COMMENTS – Zoning Division

E. Whether there are other existing or changing conditions affecting the use and development of the property which give supporting grounds for either approval or disapproval of the zoning proposal;

It is Staff's opinion that there are existing and changing conditions affecting the use and development of the property which give supporting grounds for approving the applicant's rezoning proposal if the density is limited to five (5) units per acre. The applicant has proposed to include the increased buffers adjacent to the lower density residential zoning to the east of the development. If developed, the applicant's proposal would meet the objectives of the Comprehensive Plan for the MDR future land use category and fall within the surrounding development density range.

The staff analysis and recommendation made by the Planning and Zoning Staff are only the opinions of the Planning and Zoning Staff and are by no means the final decision. The Cobb County Board of Commissioners makes the final decisions on all Rezoning and Land Use Permits at an advertised public hearing.

Summary of Intent for Rezoning*

.....
Part 1. Residential Rezoning Information (attach additional information if needed)

- a) Proposed unit square-footage(s): Minimum 1,600 square feet and greater
- b) Proposed building architecture: Traditional/Craftsman - Fronts of brick, stone, shake, board and batten, cementitious lap siding
- c) List all requested variances: See attached Listing of Variances
- _____
- _____
- _____

.....
Part 2. Non-residential Rezoning Information (attach additional information if needed)

- a) Proposed use(s): Not Applicable.
- b) Proposed building architecture: Not Applicable.
- c) Proposed hours/days of operation: Not Applicable.
- d) List all requested variances: _____
- _____
- _____
- _____

.....
Part 3. Other Pertinent Information (List or attach additional information if needed)

.....
Part 4. Is any of the property included on the proposed site plan owned by the Local, State, or Federal Government?

(Please list all Right-of-Ways, Government owned lots, County owned parcels and/or remnants, etc., and attach a plat clearly showing where these properties are located).

Yes. Portions of two Cobb County rights-of-way are proposed for abandonment, as shown on submitted Site Plan.

*Applicant specifically reserves the right to amend any information set forth in this Summary of Intent for Rezoning, or any portion of the Application for Rezoning or documentation attached thereto, at any time during the rezoning process.

ATTACHMENT TO SUMMARY OF INTENT FOR REZONING

Application No.: Z-_____ (2026)
Hearing Dates: February 3, 2026
February 17, 2026

Applicant: The Revive Land Group, LLC
Titleholder: Estate of Lucille Garrett
(a/k/a Leathy Lucille Garrett;
a/k/a Leathy Lucille Price Garrett)

Tax Parcel Identification No.: 19069100010

Part 1. Residential Rezoning Information

(c) List all requested variances:

- i) Sec. 134-201.2.(4)a. Minimum lot size: 7,000 sf. Variance request for minimum lot size: 3,700 sf;
- ii) Sec. 134-201.2.(4)b. Minimum lot width at front setback line: 70 feet; cul-de-sac or interior lot, 50 feet. Variance request for minimum lot width of 41 feet;
- iii) Sec. 134-201.2.(4)c. Minimum width between buildings: 15 feet. Variance request for minimum width between buildings of 10 feet;
- iv) Sec. 134-201.2.(4)d. Minimum public road frontage: 70 feet; cul-de-sac 35 feet; 50 feet minimum public road frontage if interior to development. Variance request for minimum public road frontage of 41 feet;
- v) Sec. 134-201.2.(4)e. Minimum building setbacks: minimum front yard setback: 20 feet. Variance request for minimum front yard setback of 15 feet;
- vi) Sec. 134-201.2.(4)e. Minimum building setbacks: minimum major side yard setback: 20 feet. Variance request for minimum front yard setback of 15 feet.
- vii) Sec. 134-201.2.(4)e. Minimum building setbacks: minimum rear yard setback on an arterial street: 40 feet. Variance request for minimum rear yard setback on an arterial street of 30 feet;
- viii) Sec. 134-201.2.(4)e. Minimum building setbacks: minimum rear yard setback: 30 feet. Variance request for minimum front yard setback of 20 feet; and
- ix) Sec. 134-191. Summary of bulk regulations: maximum coverage: 35 percent. Variance request for maximum coverage of 45percent.

Cobb County Development Standards Sec 402.7.1

- x) A minimum 50 feet straight line distance is required between residential driveways. Request a minimum 12 feet straight line distance between residential driveways; and
- xi) A minimum 50 feet straight line distance is required between the intersection radius and the edge of residential driveway aprons. Request a minimum 2 feet straight line distance between the intersection radius and the edge of the residential driveway apron.

**TRAFFIC IMPACT STUDY
FOR
PROPOSED SOMERSET AT MACEDONIA RESIDENTIAL
DEVELOPMENT ON ERNEST W BARRETT PARKWAY
COBB COUNTY, GEORGIA**



Prepared for:

***The Revive Land Group
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November 17, 2025
A & R Project # 25-196

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1.0 INTRODUCTION

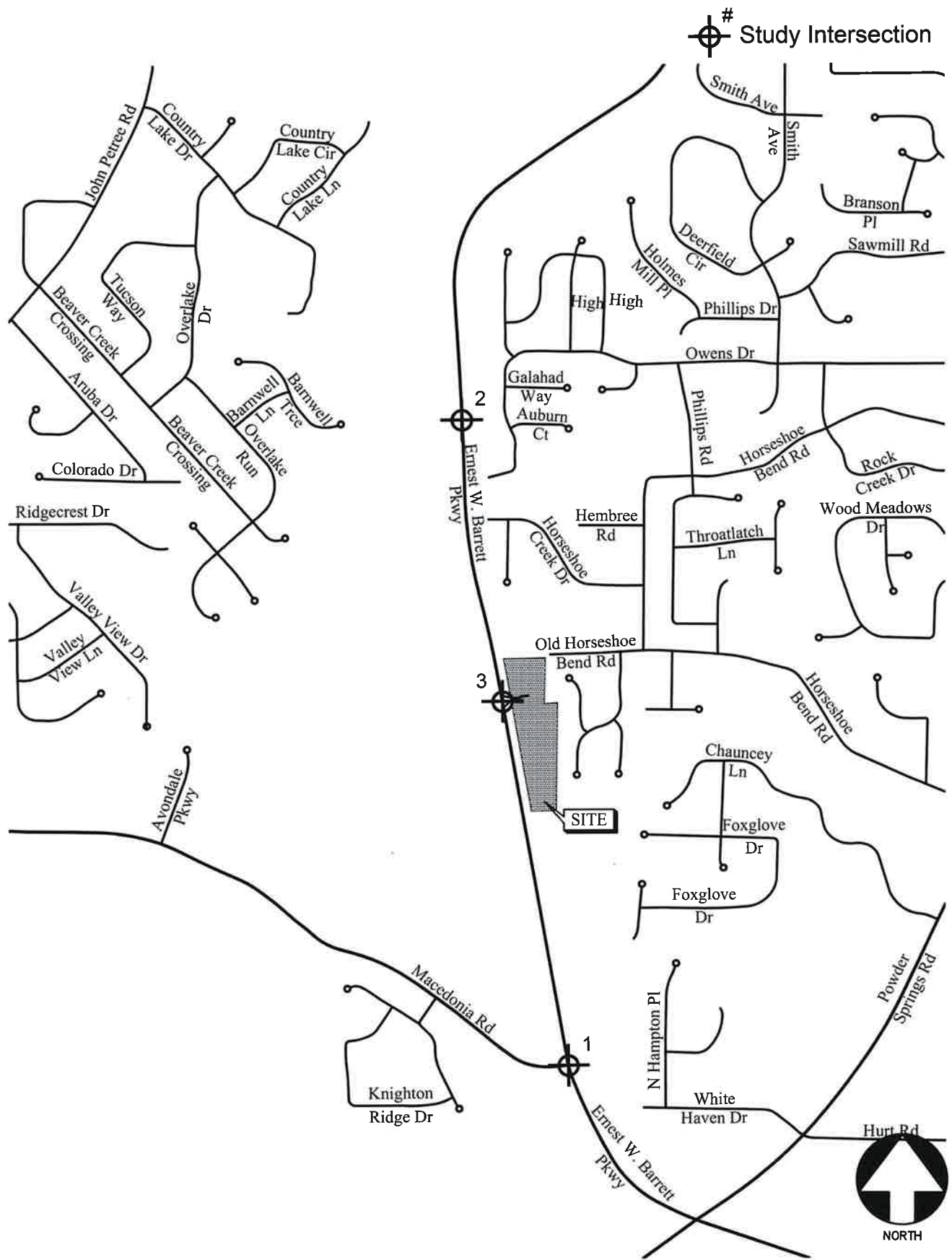
The purpose of this study is to determine the traffic impact from the proposed Somerset at Macedonia residential development that will be located on the east side of Ernest W Barrett Parkway about 3,000 feet north of Macedonia Road in Cobb County. The traffic analysis includes evaluation of the current operations as well as future conditions including traffic generated by the proposed development. The development will consist of 43 single-family detached houses and proposes one right-in/right-out driveway on Ernest W Barrett Parkway.



The AM and PM peak hours have been analyzed in this study. In addition to the site access point, this study includes the evaluation of traffic operations at the intersections of:

1. Ernest W Barrett Parkway at Macedonia Road
2. Ernest W Barrett Parkway at Median Break (approximately 0.45 miles to the north of the proposed site driveway location)

Recommendations to improve traffic operations have been identified as appropriate and are discussed in detail in the following sections of the report. The location of the development and the surrounding roadway network are shown in Figure 1.



LOCATION MAP

FIGURE 1

A&R Engineering Inc.

2.0 EXISTING FACILITIES / CONDITIONS

2.1 Roadway Facilities

The following is a brief description of each of the roadway facilities located in proximity to the site:

2.1.1 Ernest W Barrett Parkway

Ernest W Barrett Parkway is a north-south, four-lane, median divided roadway with a posted speed limit of 45 mph in the vicinity of the site. Georgia Department of Transportation (GDOT) traffic counts (Station ID: 067-0657) indicate that the daily traffic volume on Ernest W Barrett Parkway in 2024 was 21,500 vehicles per day northwest of Macedonia Road. Cobb County DOT classifies Ernest W Barrett Parkway as an arterial roadway.

2.1.2 Macedonia Road

Macedonia Road is an east-west, two-lane, undivided roadway with a two way left turn lane and posted speed limit of 35 mph in the vicinity of the site. Georgia Department of Transportation (GDOT) traffic counts (Station ID: 067-8836) indicate that the daily traffic volume on Macedonia Road in 2024 was 5,660 vehicles per day southwest of Hopkins Road. Cobb County DOT classifies Macedonia Road as a major collector roadway.

3.0 STUDY METHODOLOGY

In this study, the methodology used for evaluating traffic operations at each of the subject intersections is based on the criteria set forth in the Transportation Research Board's Highway Capacity Manual, 6th edition (HCM 6). Synchro software, which utilizes the HCM methodology, was used for the analysis. The following is a description of the methodology employed for the analysis of unsignalized and signalized intersections.

3.1 Unsignalized Intersections

For unsignalized intersections controlled by a stop sign on minor streets, the level of service (LOS) for motor vehicles with controlled movements is determined by the computed control delay according to the thresholds stated in Table 1 below. LOS is determined for each minor street movement (or shared movement), as well as major street left turns. LOS is not defined for the intersection as a whole or for major street approaches. The LOS of any controlled movement which experiences a volume to capacity ratio greater than 1 is designated as "F" regardless of the control delay.

Control delay for unsignalized intersections includes initial deceleration delay, queue move-up time, stopped delay, and final acceleration delay. Several factors affect the control delay for unsignalized intersections, such as the availability and distribution of gaps in the conflicting traffic stream, critical gaps, and follow-up time for a vehicle in the queue.

Level of service is assigned a letter designation from "A" through "F". Level of service "A" indicates excellent operations with little delay to motorists, while level of service "F" exists when there are insufficient gaps of acceptable size to allow vehicles on the side street to cross the main road without experiencing long delays.

TABLE 1 — LEVEL OF SERVICE CRITERIA FOR UNSIGNALIZED INTERSECTIONS		
Control Delay (sec/vehicle)	LOS by Volume-to-Capacity Ratio*	
	$v/c \leq 1.0$	$v/c > 1.0$
≤ 10	A	F
> 10 and ≤ 15	B	F
> 15 and ≤ 25	C	F
> 25 and ≤ 35	D	F
> 35 and ≤ 50	E	F
> 50	F	F

*The LOS criteria apply to each lane on a given approach and to each approach on the minor street. LOS is not calculated for major-street approaches or for the intersection.

Source: Highway Capacity Manual, 6th edition, Exhibit 20-2 *LOS Criteria: Motorized Vehicle Mode*

3.2 Signalized Intersections

According to HCM procedures, LOS can be calculated for the entire intersection, each intersection approach, and each lane group. HCM uses control delay alone to characterize LOS for the entire intersection or an approach. Control delay per vehicle is composed of initial deceleration delay, queue move-up time, stopped delay, and final acceleration delay. Both control delay and volume-to-capacity ratio are used to characterize LOS for a lane group. A volume-to-capacity ratio of greater than 1.0 for a lane group indicates failure from capacity perspective. Therefore, such a lane group is assigned LOS F regardless of the amount of control delay.

Table 2 below summarizes the LOS criteria from HCM for motorized vehicles at signalized intersection.

TABLE 2 — LEVEL OF SERVICE CRITERIA FOR SIGNALIZED INTERSECTIONS		
Control Delay (sec/vehicle) *	LOS for Lane Group by Volume-to-Capacity Ratio*	
	v/c ≤ 1.0	v/c > 1.0
≤ 10	A	F
> 10 and ≤ 20	B	F
> 20 and ≤ 35	C	F
> 35 and ≤ 55	D	F
> 55 and ≤ 80	E	F
> 80	F	F

*For approach-based and intersection wide assessments, LOS is defined solely by control delay

Source: Highway Capacity Manual, 6th edition, Exhibit 19-8 *LOS Criteria: Motorized Vehicle Mode*

LOS A is typically assigned when the volume-to-capacity (v/c) ratio is low and either progression is exceptionally favourable, or the cycle length is very short. LOS B is typically assigned when the v/c ratio is low and either progression is highly favorable, or the cycle length is short. However, more vehicles are stopped than with LOS A. LOS C is typically assigned when progression is favorable, or the cycle length is moderate. Individual *cycle failures* (one or more queued vehicles are not able to depart because of insufficient capacity during the cycle) may begin to appear at this level. Many vehicles still pass through the intersection without stopping, but the number of vehicles stopping is significant. LOS D is typically assigned when the v/c ratio is high and either progression is ineffective, or the cycle length is long. There are many vehicle-stops and individual cycle failures are noticeable. LOS E is typically assigned when the v/c ratio is high, progression is very poor, the cycle length is long, and individual cycle failures are frequent. LOS F is typically assigned when the v/c ratio is very high, progression is very poor, the cycle length is long, and most cycles fail to clear the queue.

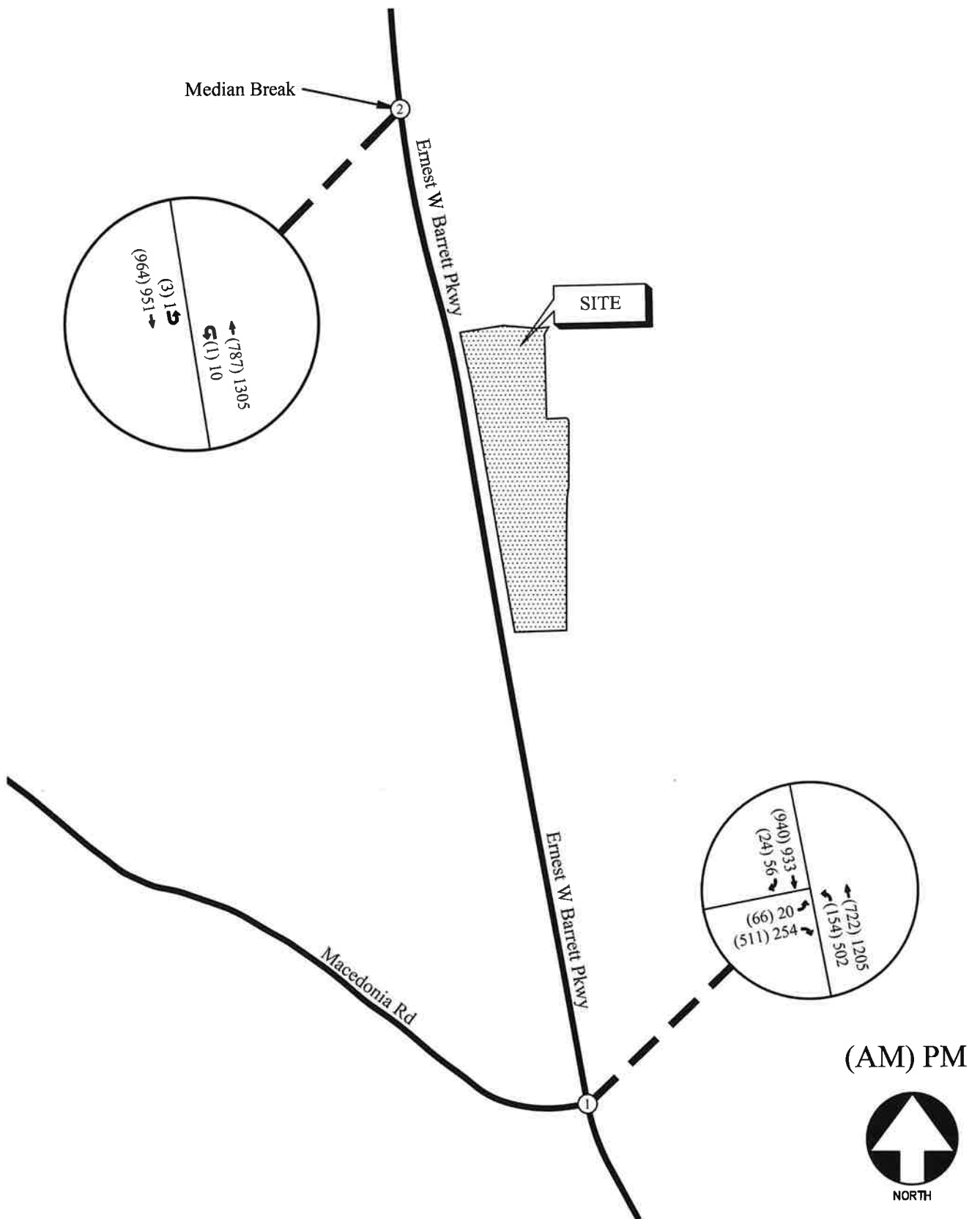
4.0 EXISTING 2025 TRAFFIC ANALYSIS

4.1 Existing Traffic Volumes

Existing traffic counts were obtained at the following study intersections:

1. Ernest W Barrett Parkway at Macedonia Road
2. Ernest W Barrett Parkway at Median Break

Turning movement counts were collected on Wednesday, October 22, 2025. All turning movement counts were recorded during the AM and PM peak hours between 7:00am to 9:00am and 4:00pm to 6:00pm, respectively. The four consecutive 15-minute interval volumes that summed to produce the highest volume at the intersections were then determined. These volumes make up the peak hour traffic volumes for the intersections and are shown in Figure 2. The existing traffic control and lane geometry for the intersections are shown in Figure 3.

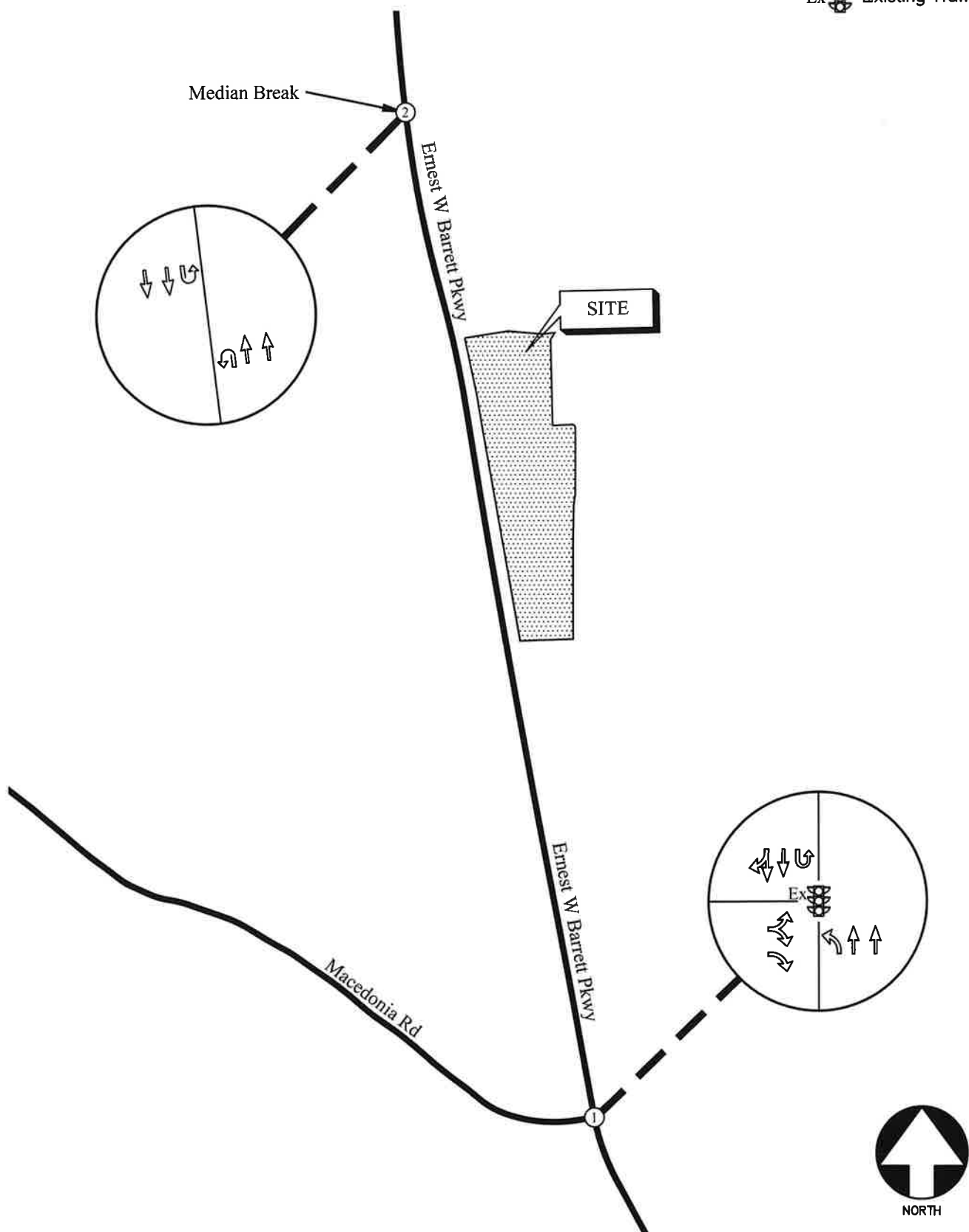


EXISTING WEEKDAY PEAK-HOUR VOLUMES

FIGURE 2
A&R Engineering Inc.

LEGEND

- Ex  Existing Signed Approach
-  Existing Lane Geometry
- Ex  Existing Traffic Signal



EXISTING TRAFFIC CONTROL AND LANE GEOMETRY

FIGURE 3
A&R Engineering Inc.

Figure 3 – Existing Traffic Control and Lane Geometry

4.2 Existing Traffic Operations

Existing 2025 traffic operations were analyzed at the study intersections in accordance with the HCM methodology. The results of the analyses are shown in Table 3.

TABLE 3 — EXISTING INTERSECTION OPERATIONS				
Intersection		Traffic Control	LOS (Delay)	
			AM Peak Hour	PM Peak Hour
1	<u>Ernest W Barrett Parkway @ Macedonia Road</u>	Signalized	<u>C (23.0)</u>	<u>C (21.0)</u>
	-Eastbound Approach		D (54.9)	E (74.4)
	-Northbound Approach		A (8.8)	B (15.2)
	-Southbound Approach		B (16.6)	B (16.0)
2	<u>Ernest W Barrett Parkway @ Median Break</u>	-	A (0.0)	A (0.0)
	-Northbound U-Turn		A (0.0)	A (0.0)
	-Southbound U-Turn		A (0.0)	A (0.0)

The results of the existing traffic operations analysis indicate that the signalized study intersection is operating at an overall level of service “B” in both the AM and PM peak hours.

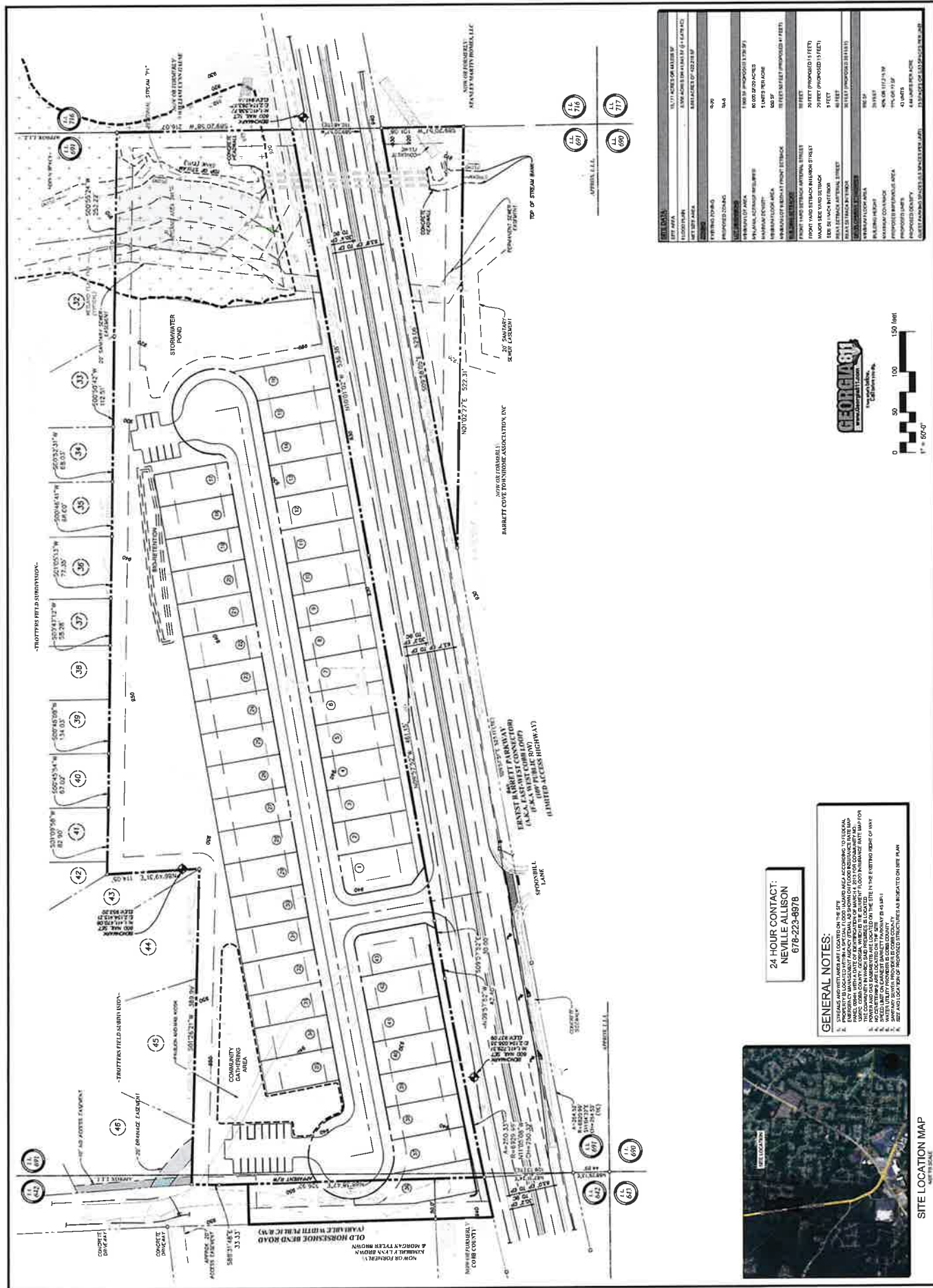
5.0 PROPOSED DEVELOPMENT

The proposed development is the Somerset at Macedonia residential development that will be located on the east side of Ernest W Barrett Parkway about 3,000 feet north of Macedonia Road in Cobb County. The development will consist of 43 single-family detached houses and proposes one right-in/right-out driveway on Ernest W Barrett Parkway.



A site plan is shown in Figure 4.

Figure 4 – Site Plan



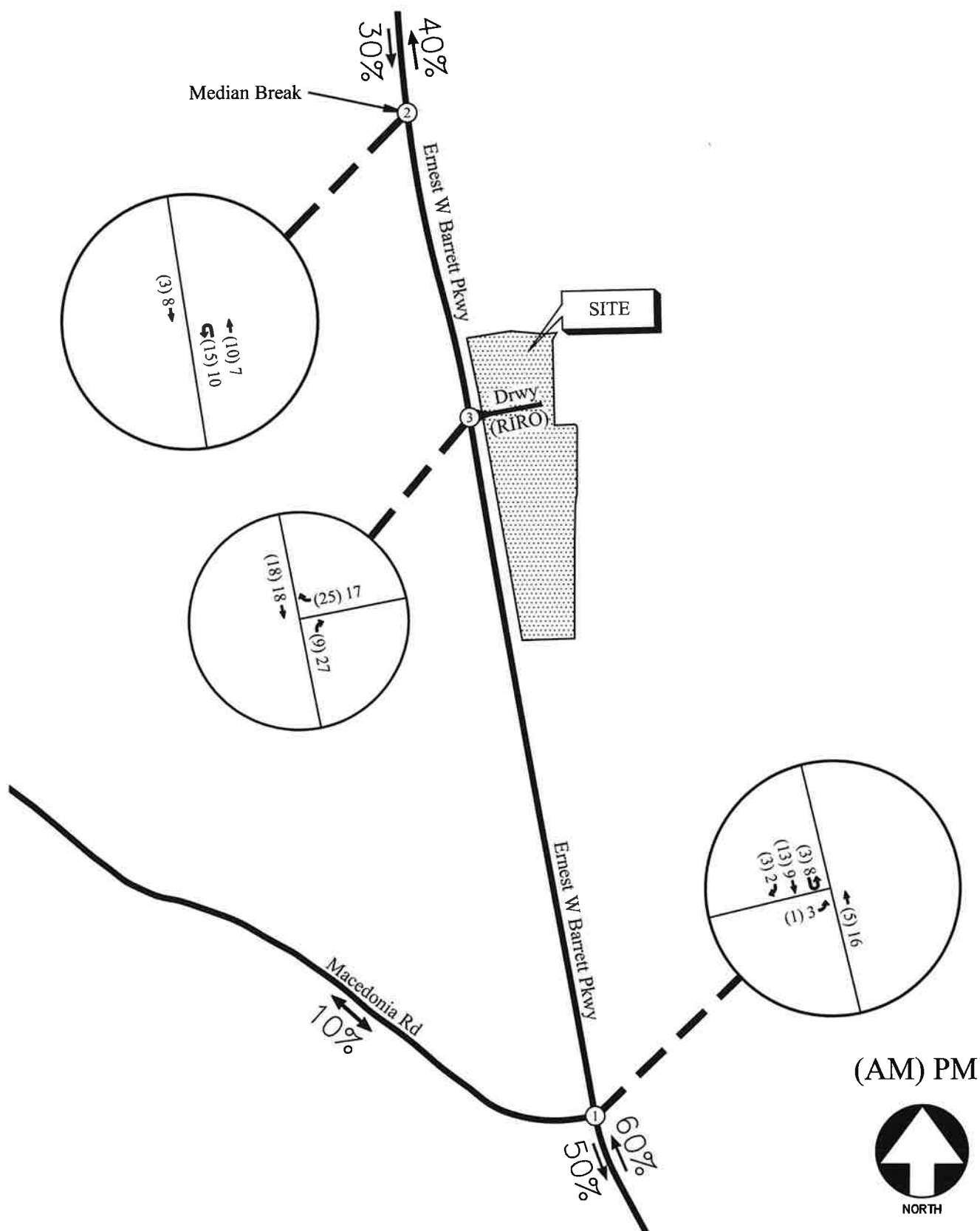
5.1 Trip Generation

Trip generation estimates for the project were based on the rates and equations published in the 12th edition of the Institute of Transportation Engineers (ITE) Trip Generation Manual. This reference contains traffic volume count data collected at similar facilities nationwide. The trip generation was based on the following ITE Land Use category: *210 – Single-Family Detached Housing*. The calculated trip generation volumes for the proposed development are shown in Table 4.

TABLE 4 – TRIP GENERATION FOR PROPOSED DEVELOPMENT								
Land Use	Size	AM Peak Hour			PM Peak Hour			24 Hour
		Enter	Exit	Total	Enter	Exit	Total	2-way
ITE 210 – Single-Family Detached Housing	43 Units	9	25	34	27	17	44	612

5.2 Trip Distribution

The trip distribution describes how traffic arrives and departs from the site. An overall trip distribution was developed for the site based on a review of the existing travel patterns in the area and the locations of major roadways and highways that will serve the development. The site-generated peak hour traffic volumes, shown in Table 4, were assigned to the study area intersections based on this distribution. The outer-leg distribution and AM and PM peak hour traffic volumes generated by the site are shown in Figure 5.



TRIP DISTRIBUTION AND NEW SITE-GENERATED
WEEKDAY PEAK HOUR VOLUMES

FIGURE 5
A&R Engineering Inc.

6.0 FUTURE 2027 TRAFFIC ANALYSIS

The future 2027 traffic operations are analyzed for the “Build” and “No-Build” conditions.

6.1 Future “No-Build” Conditions

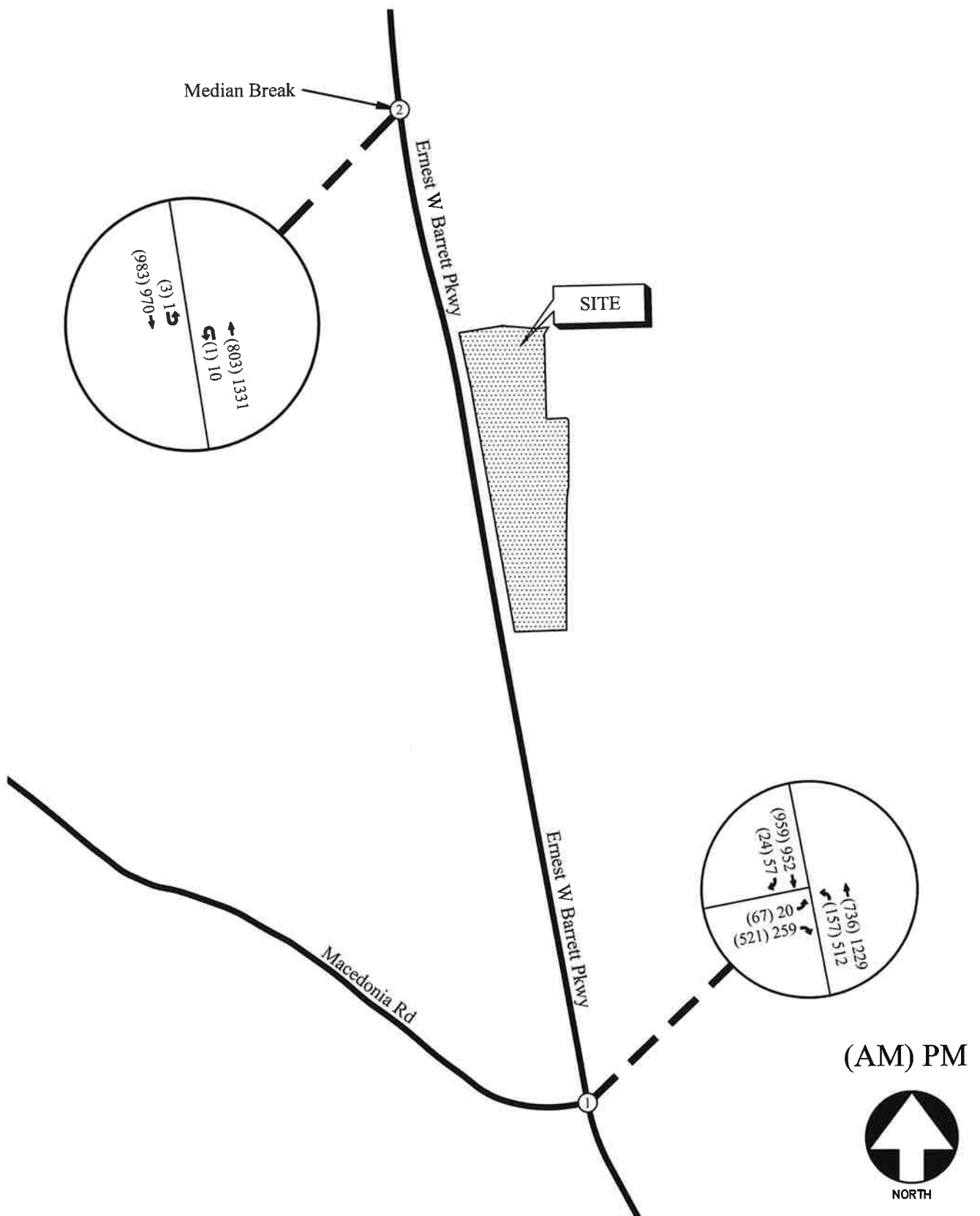
The “No-Build” (or background) conditions provide an assessment of how traffic will operate in the study horizon year without the study site being developed as proposed, with projected increases in through traffic volumes due to normal annual growth. The Future “No-Build” volumes consist of the existing traffic volumes (Figure 2) plus increases for annual growth of through traffic.

6.1.1 Annual Traffic Growth

In order to evaluate future traffic operations in this area, a projection of normal traffic growth was applied to the existing volumes. The Georgia Department of Transportation recorded average daily traffic volumes at several locations in the vicinity of the site. Reviewing the growth over the last three (2022-2024) years revealed growth of approximately 1% in the area. This growth factor was applied to the existing traffic volumes between collector and arterial roadways to estimate the future year traffic volumes prior to the addition of site-generated traffic. The resulting future “No-Build” volumes on the roadway are shown in Figure 6.

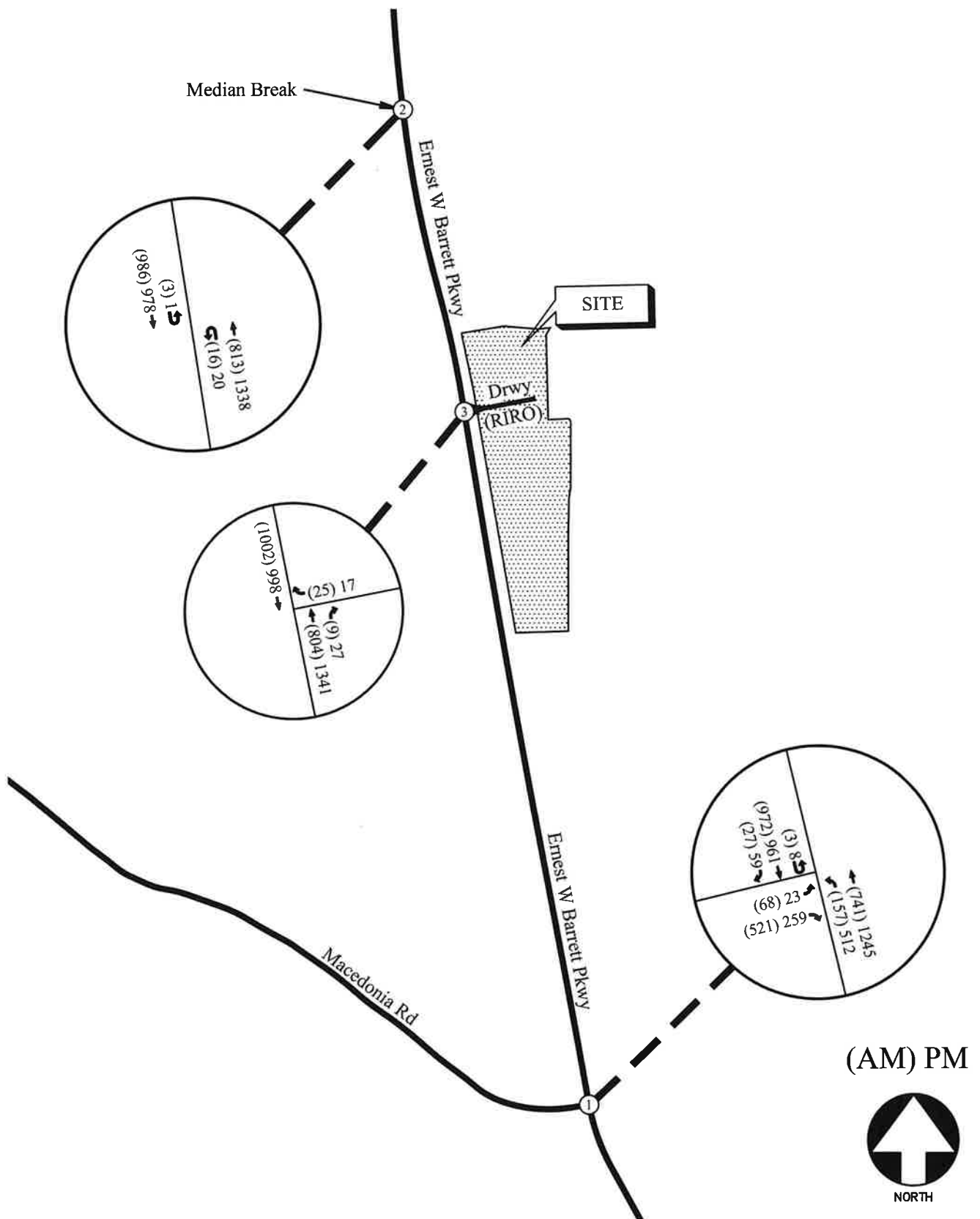
6.2 Future “Build” Conditions

The “Build” or development conditions include the estimated background traffic from the “No-Build” conditions plus the added traffic from the proposed development. To evaluate future traffic operations in this area, the additional traffic volumes from the site (Figure 5) were added to base traffic volumes (Figure 6) to calculate the future traffic volumes after the construction of the development. These total future “Build” traffic volumes are shown in Figure 7.



FUTURE (NO-BUILD) WEEKDAY PEAK HOUR VOLUMES

FIGURE 6
A&R Engineering Inc.



FUTURE (BUILD) WEEKDAY PEAK HOUR VOLUMES

FIGURE 7
A&R Engineering Inc.

6.3 Auxiliary Lane Analysis

Included below is the analysis for a right turn lane for the site driveway (per Cobb County Standards). The analyses below are based off the trip distribution included in Section 5.2. According to the trip distribution, the 24-hour two-way volume for traffic entering and exiting the site are 612 vehicles.

6.3.1 Deceleration Turn Lane Analysis

Per Cobb County Development Standards, (402.9), right turn lanes are required on any roadway that is included in the county's major thoroughfare plan network. Since Ernest W Barrett Parkway is an arterial roadway and is included in Cobb County's Major Thoroughfare Plan, a right turn lane is required at site driveway.

6.4 Future Traffic Operations

The future “No-Build” and “Build” traffic operations were analysed using the volumes in Figures 6 and 7, respectively. The results of the future traffic operations analysis are shown below in Table 5. Recommendations on traffic control and lane geometry are shown in Figure 8.

TABLE 5 – FUTURE INTERSECTION OPERATIONS					
Intersection		LOS (Delay)			
		NO BUILD		BUILD	
		AM Peak	PM Peak	AM Peak	PM Peak
1	<u>Ernest W Barrett Parkway @ Macedonia Road</u>				
	-Eastbound Approach	<u>C (23.2)</u>	<u>C (22.4)</u>	<u>C (23.3)</u>	<u>C (22.7)</u>
	-Northbound Approach	D (54.2)	E (74.6)	D (54.3)	E (74.8)
	-Southbound Approach	A (9.2)	B (16.5)	A (9.2)	B (16.8)
2	<u>Ernest W Barrett Parkway @ Median Break</u>				
	-Northbound U-Turn	B (17.3)	B (17.9)	B (17.5)	B (18.5)
	-Southbound U-Turn				
3	<u>Ernest W Barrett Parkway @ Site Driveway</u>				
	<u>(Right In/Right Out)</u>				
	-Westbound Approach				

The results of the future “No-Build” and “Build” traffic operations analysis indicate that the unsignalized study intersections will continue to operate at a level of service “C” or better in both the AM and PM peak hours. The signalized study intersection will be operating at a level of service “C” or better in both the AM and PM peak hour.

Table 6 below includes 95th percentile SimTraffic queue lengths at the study intersections. SimTraffic results of the “No Build” and “Build” have been used since the microsimulation more accurately modelled a multi-lane turn lane scenario than Synchro (macrosimulation).

TABLE 6 – FUTURE INTERSECTION 95 TH PERCENTILE QUEUES						
Intersection		Available Storage	Queue Length in feet			
			NO-BUILD		BUILD	
			AM Peak	PM Peak	AM Peak	PM Peak
1	<u>Ernest W Barrett Parkway @ Macedonia Road</u>					
	-Eastbound Left/Right	460'	227	94	229	97
	-Eastbound Right	-	76	73	76	73
	-Northbound Left	305'	77	454	78	464
	-Northbound Through	-	172	183	174	190
	-Southbound U-Turn	190'	-	-	7	16
	-Southbound Through	-	411	504	419	511
	-Southbound Through/ Right	-	411	504	419	511
2	<u>Ernest W Barrett Parkway @ Median Break</u>					
	-Northbound U-Turn	145'	4	23	35	32
	-Southbound U-Turn	155'	11	6	12	5

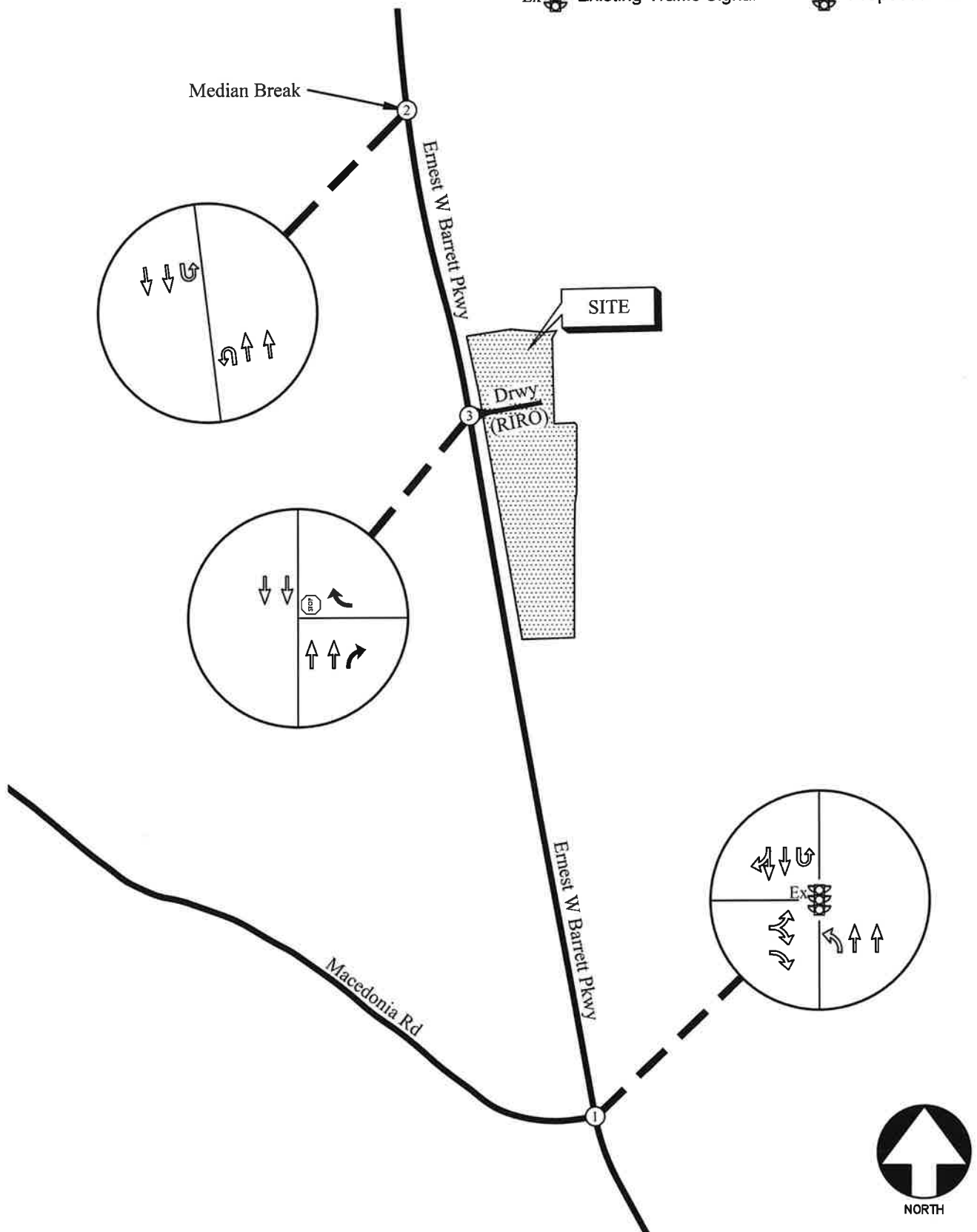
The storage lengths are all adequate to accommodate projected queues other than the northbound left approach at the intersection of Ernest W Barrett Parkway and Macedonia Road during the PM peak hour.

6.5 System Improvements

The “No-Build” scenario shows a significant amount of northbound left turning volume at Ernest W Barrett Parkway and Macedonia Road during the PM peak hour. The expected 95th percentile queue during this time is also longer than the available storage for this movement. It is recommended to consider a dual left turn lane whenever the left turn volume is greater than 300 vehicles during the peak hour. The PM peak hour volume for northbound left turning vehicles at this intersection is 512.

LEGEND

- | | | | |
|--|--------------------------|--|--------------------------|
| Ex  | Existing Signed Approach |  (stop) | Proposed Signed Approach |
|  | Existing Lane Geometry |  | Proposed Lane Geometry |
| Ex  | Existing Traffic Signal |  | Proposed Traffic Signal |



FUTURE TRAFFIC CONTROL AND LANE GEOMETRY

FIGURE 8
A&R Engineering Inc.

7.0 CONCLUSIONS AND RECOMMENDATIONS

Traffic impacts were evaluated for the proposed Somerset at Macedonia residential development that will be located on the east side of Ernest W Barrett Parkway in Cobb County. The development will consist of 43 single-family detached houses and proposes one right-in/right-out driveway on Ernest W Barrett Parkway.

Existing and future operations after completion of the project were analyzed at the intersections of:

1. Ernest W Barrett Parkway at Macedonia Road
2. Ernest W Barrett Parkway at Median Break (Approximately 0.45 miles to the north of the proposed site driveway location)
3. Ernest W Barrett Parkway at Site Driveway (Right In/Right Out)

The analysis included the evaluation of future operations for “No-Build” and “Build” conditions, with the differences between “No-Build” and “Build” accounting for an increase in traffic due to the proposed development. The results of the future “No-Build” and “Build” traffic operations analysis indicate that the unsignalized study intersections will continue to operate at a level of service “C” in both the AM and PM peak hours. The results of the future “No-Build” and “Build” traffic operations analysis also indicate that the signalized study intersections will continue to operate at a level of service “C” in both the AM and PM peak hours. The impact of the proposed site’s traffic on the traffic operations in the study network is minimal.

7.1 Recommendation for Site Access Configuration

The following access configuration is recommended for the proposed site driveway intersection.

- Site Driveway: Right-In/ Right-Out driveway on Ernest W Barrett Parkway
 - One entering and one exiting lanes.
 - Stop-sign controlled on the driveway approach with Ernest W Barrett Parkway remaining free flow.
 - Deceleration turn lane for entering traffic
 - Provide adequate sight distance per AASHTO standards.

Appendix

Existing Intersection Traffic Counts
Linear Regression of Daily Traffic.....
Existing Intersection Analysis.....
Future “No-Build” Intersection Analysis.....
Future “Build” Intersection Analysis.....
Traffic Volume Worksheets

EXISTING INTERSECTION TRAFFIC COUNTS

2160 Kingston Court Suite 'O'
Marietta, GA 30067

File Name : 20250374
Site Code : 20250374
Start Date : 10-22-2025
Page No : 1

Groups Printed- Cars, Buses & Trucks																				
	Ernest W Barrett Pkwy Northbound					Ernest W Barrett Pkwy Southbound				Eastbound				Westbound						
Start Time	Left	Thru	Right	U-Turn	App. Total	Left	Thru	Right	U-Turn	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Int. Total	
07:00 AM	0	191	0	0	191	0	208	0	2	210	0	0	0	0	0	0	0	0	0	401
07:15 AM	0	209	0	0	209	0	258	0	0	258	0	0	0	0	0	0	0	0	0	467
07:30 AM	0	204	0	0	204	0	265	0	0	265	0	0	0	0	0	0	0	0	0	469
07:45 AM	0	183	0	1	184	0	233	0	1	234	0	0	0	0	0	0	0	0	0	418
Total	0	787	0	1	788	0	964	0	3	967	0	0	0	0	0	0	0	0	0	1755
08:00 AM	0	169	0	1	170	0	228	0	1	229	0	0	0	0	0	0	0	0	0	399
08:15 AM	0	168	0	2	170	0	232	0	0	232	0	0	0	0	0	0	0	0	0	402
08:30 AM	0	147	0	1	148	0	221	0	2	223	0	0	0	0	0	0	0	0	0	371
08:45 AM	0	172	0	0	172	0	263	0	0	263	0	0	0	0	0	0	0	0	0	435
Total	0	656	0	4	660	0	944	0	3	947	0	0	0	0	0	0	0	0	0	1607
*** BREAK ***																				
04:00 PM	0	308	0	2	310	0	232	0	0	232	0	0	0	0	0	0	0	0	0	542
04:15 PM	0	321	0	3	324	0	216	0	0	216	0	0	0	0	0	0	0	0	0	540
04:30 PM	0	323	0	2	325	0	230	0	0	230	0	0	0	0	0	0	0	0	0	555
04:45 PM	0	354	0	3	357	0	248	0	1	249	0	0	0	0	0	0	0	0	0	606
Total	0	1306	0	10	1316	0	926	0	1	927	0	0	0	0	0	0	0	0	0	2243
05:00 PM	0	307	0	2	309	0	257	0	0	257	0	0	0	0	0	0	0	0	0	566
05:15 PM	0	290	0	2	292	0	222	0	1	223	0	0	0	0	0	0	0	0	0	515
05:30 PM	0	291	0	2	293	0	267	0	2	269	0	0	0	0	0	0	0	0	0	562
05:45 PM	0	280	0	2	282	0	240	0	1	241	0	0	0	0	0	0	0	0	0	523
Total	0	1168	0	8	1176	0	986	0	4	990	0	0	0	0	0	0	0	0	0	2166
Grand Total	0	3917	0	23	3940	0	3820	0	11	3831	0	0	0	0	0	0	0	0	0	7771
Apprch %	0	99.4	0	0.6		0	99.7	0	0.3		0	0	0		0	0	0			
Total %	0	50.4	0	0.3	50.7	0	49.2	0	0.1	49.3	0	0	0	0	0	0	0	0	0	

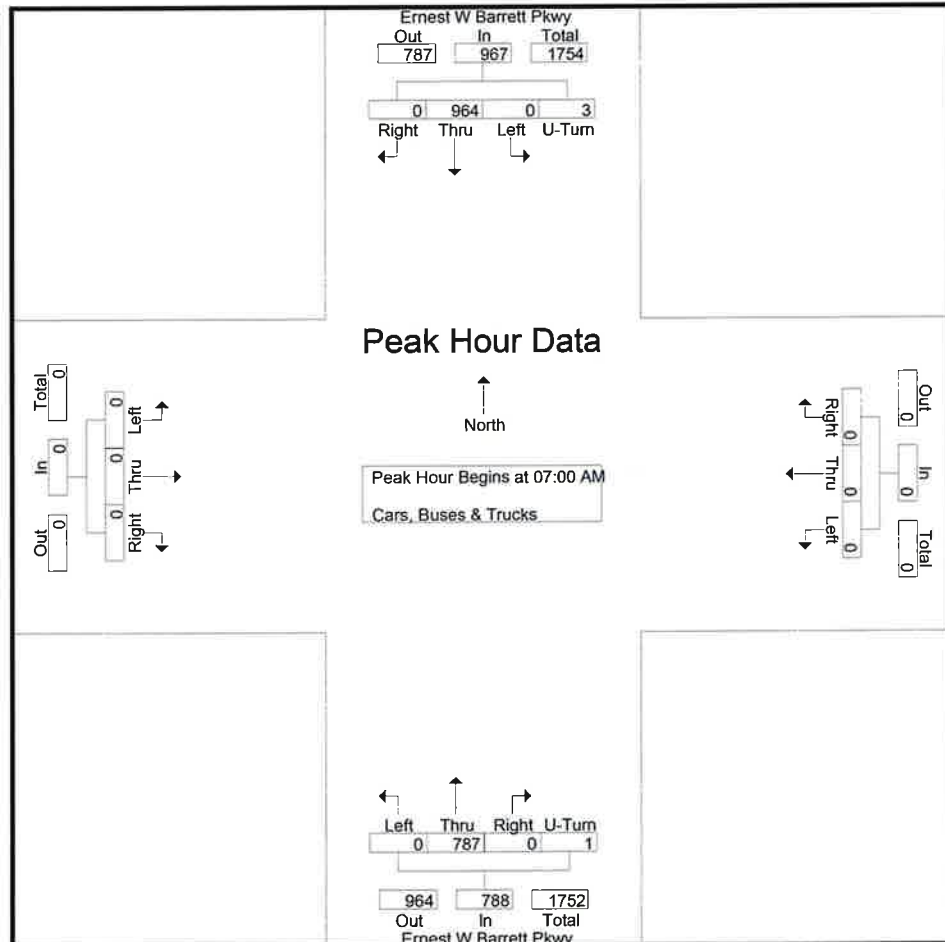
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Marietta, GA 30067

TMC Data
Ernest W Barrett Pkwy @ Median Opening
North of Macedonia Road
7-9 am | 4-6 pm

File Name : 20250374
Site Code : 20250374
Start Date : 10-22-2025
Page No : 2

	Ernest W Barrett Pkwy Northbound					Ernest W Barrett Pkwy Southbound					Eastbound				Westbound				
Start Time	Left	Thru	Right	U-Turn	App. Total	Left	Thru	Right	U-Turn	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	In/L Total
Peak Hour Analysis From 07:00 AM to 08:45 AM - Peak 1 of 1																			
Peak Hour for Entire Intersection Begins at 07:00 AM																			
07:00 AM	0	191	0	0	191	0	208	0	2	210	0	0	0	0	0	0	0	0	401
07:15 AM	0	209	0	0	209	0	258	0	0	258	0	0	0	0	0	0	0	0	467
07:30 AM	0	204	0	0	204	0	265	0	0	265	0	0	0	0	0	0	0	0	469
07:45 AM	0	183	0	1	184	0	233	0	1	234	0	0	0	0	0	0	0	0	418
Total Volume	0	787	0	1	788	0	964	0	3	967	0	0	0	0	0	0	0	0	1755
% App. Total	0	99.9	0	0.1		0	99.7	0	0.3		0	0	0		0	0	0		
PHF	.000	.941	.000	.250	.943	.000	.909	.000	.375	.912	.000	.000	.000	.000	.000	.000	.000	.000	.936



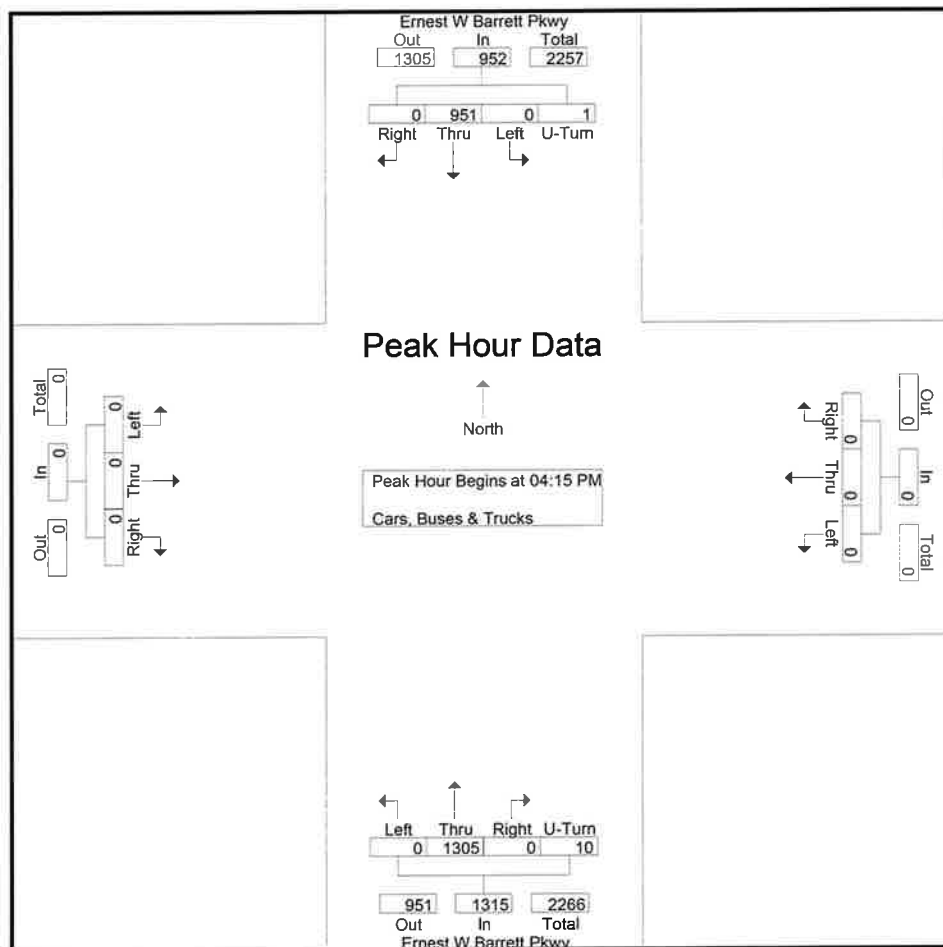
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TMC Data
Ernest W Barrett Pkwy @ Median Opening
North of Macedonia Road
7-9 am | 4-6 pm

File Name : 20250374
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Start Date : 10-22-2025
Page No : 3

	Ernest W Barrett Pkwy Northbound					Ernest W Barrett Pkwy Southbound					Eastbound				Westbound				
Start Time	Left	Thru	Right	U-Turn	App. Total	Left	Thru	Right	U-Turn	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Int. Total
Peak Hour Analysis From 04:00 PM to 05:45 PM - Peak 1 of 1																			
Peak Hour for Entire Intersection Begins at 04:15 PM																			
04:15 PM	0	321	0	3	324	0	216	0	0	216	0	0	0	0	0	0	0	0	540
04:30 PM	0	323	0	2	325	0	230	0	0	230	0	0	0	0	0	0	0	0	555
04:45 PM	0	354	0	3	357	0	248	0	1	249	0	0	0	0	0	0	0	0	606
05:00 PM	0	307	0	2	309	0	257	0	0	257	0	0	0	0	0	0	0	0	566
Total Volume	0	1305	0	10	1315	0	951	0	1	952	0	0	0	0	0	0	0	0	2267
% App. Total	0	99.2	0	0.8		0	99.9	0	0.1		0	0	0	0	0	0	0	0	
PHF	.000	.922	.000	.833	.921	.000	.925	.000	.250	.926	.000	.000	.000	.000	.000	.000	.000	.000	.935



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2160 Kingston Court Suite 'O'

Marietta, GA 30067

TMC Data

Ernest W Barrett Pkwy @ Macedonia Road

7-9 am | 4-6 pm

File Name : 20250375

Site Code : 20250375

Start Date : 10-22-2025

Page No : 1

Groups Printed- Cars, Buses & Trucks

Start Time	Ernest W Barrett Pkwy Northbound				Ernest W Barrett Pkwy Southbound				Macedonia Road Eastbound				Westbound				Int. Total
	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	
07:00 AM	12	165	0	177	0	196	12	208	26	0	151	177	0	0	0	0	562
07:15 AM	34	195	0	229	0	252	6	258	14	0	161	175	0	0	0	0	662
07:30 AM	45	190	0	235	0	265	0	265	14	0	111	125	0	0	0	0	625
07:45 AM	63	172	0	235	0	227	6	233	12	0	88	100	0	0	0	0	568
Total	154	722	0	876	0	940	24	964	66	0	511	577	0	0	0	0	2417
08:00 AM	40	160	0	200	0	223	5	228	10	0	91	101	0	0	0	0	529
08:15 AM	46	163	0	209	0	222	10	232	7	0	86	93	0	0	0	0	534
08:30 AM	32	141	0	173	0	217	4	221	7	0	72	79	0	0	0	0	473
08:45 AM	35	159	0	194	0	259	4	263	13	0	72	85	0	0	0	0	542
Total	153	623	0	776	0	921	23	944	37	0	321	358	0	0	0	0	2078
*** BREAK ***																	
04:00 PM	91	302	0	393	0	222	10	232	8	0	63	71	0	0	0	0	696
04:15 PM	88	317	0	405	0	205	11	216	7	0	57	64	0	0	0	0	685
04:30 PM	98	317	0	415	0	222	8	230	8	0	57	65	0	0	0	0	710
04:45 PM	101	353	0	454	0	233	15	248	4	0	53	57	0	0	0	0	759
Total	378	1289	0	1667	0	882	44	926	27	0	230	257	0	0	0	0	2850
05:00 PM	107	307	0	414	0	243	14	257	2	0	62	64	0	0	0	0	735
05:15 PM	130	287	0	417	0	212	10	222	5	0	63	68	0	0	0	0	707
05:30 PM	116	290	0	406	0	248	19	267	3	0	55	58	0	0	0	0	731
05:45 PM	149	321	0	470	0	230	13	243	10	0	74	84	0	0	0	0	797
Total	502	1205	0	1707	0	933	56	989	20	0	254	274	0	0	0	0	2970
Grand Total	1187	3839	0	5026	0	3676	147	3823	150	0	1316	1466	0	0	0	0	10315
Apprch %	23.6	76.4	0		0	96.2	3.8		10.2	0	89.8		0	0	0		
Total %	11.5	37.2	0	48.7	0	35.6	1.4	37.1	1.5	0	12.8	14.2	0	0	0	0	

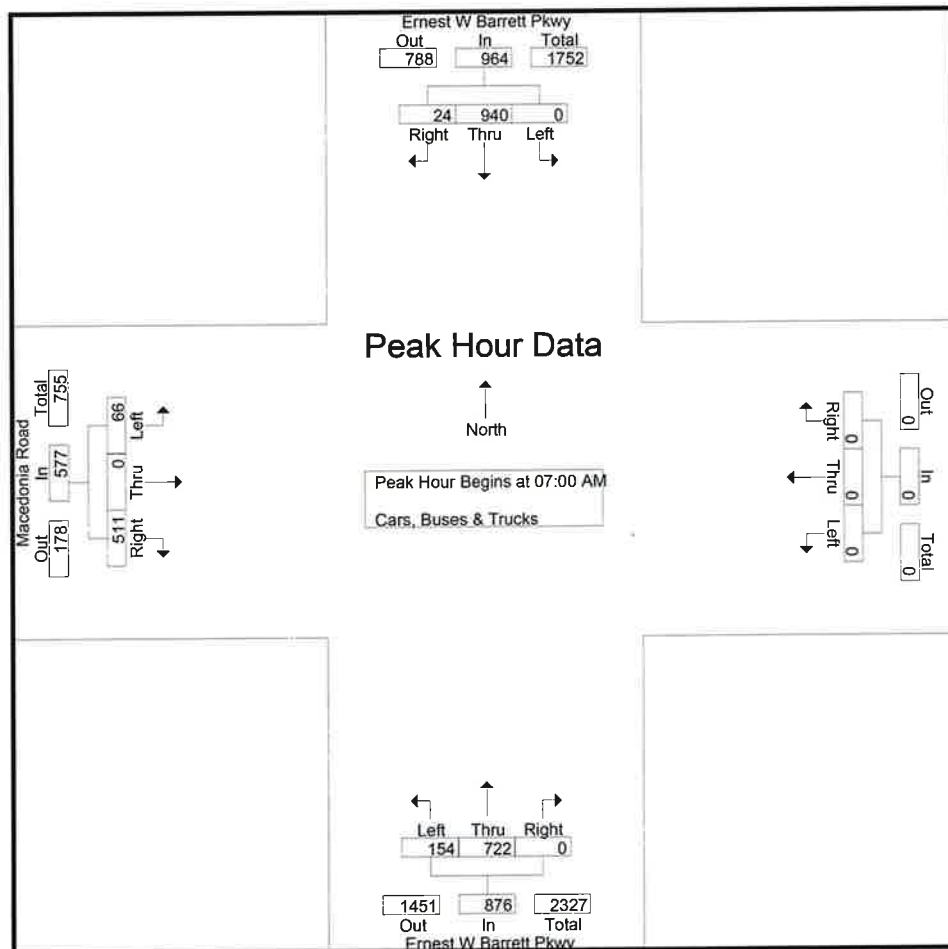
A & R Engineering, Inc.

2160 Kingston Court Suite 'O'
Marietta, GA 30067

TMC Data
Ernest W Barrett Pkwy @ Macedonia Road
7-9 am | 4-6 pm

File Name : 20250375
Site Code : 20250375
Start Date : 10-22-2025
Page No : 2

	Ernest W Barrett Pkwy Northbound				Ernest W Barrett Pkwy Southbound				Macedonia Road Eastbound				Westbound				
Start Time	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Int. Total
Peak Hour Analysis From 07:00 AM to 08:45 AM - Peak 1 of 1																	
Peak Hour for Entire Intersection Begins at 07:00 AM																	
07:00 AM	12	165	0	177	0	196	12	208	26	0	151	177	0	0	0	0	562
07:15 AM	34	195	0	229	0	252	6	258	14	0	161	175	0	0	0	0	662
07:30 AM	45	190	0	235	0	265	0	265	14	0	111	125	0	0	0	0	625
07:45 AM	63	172	0	235	0	227	6	233	12	0	88	100	0	0	0	0	568
Total Volume	154	722	0	876	0	940	24	964	66	0	511	577	0	0	0	0	2417
% App. Total	17.6	82.4	0		0	97.5	2.5		11.4	0	88.6		0	0	0		
PHF	.611	.926	.000	.932	.000	.887	.500	.909	.635	.000	.793	.815	.000	.000	.000	.000	.913



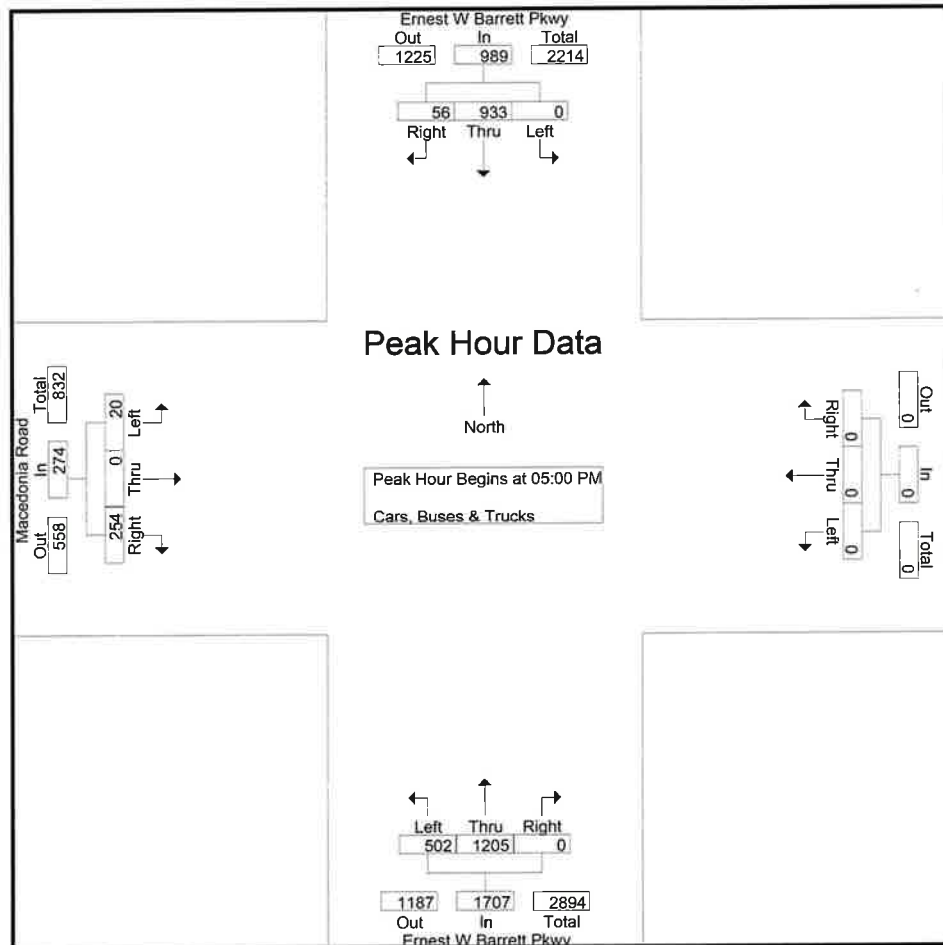
A & R Engineering, Inc.

2160 Kingston Court Suite 'O'
Marietta, GA 30067

TMC Data
Ernest W Barrett Pkwy @ Macedonia Road
7-9 am | 4-6 pm

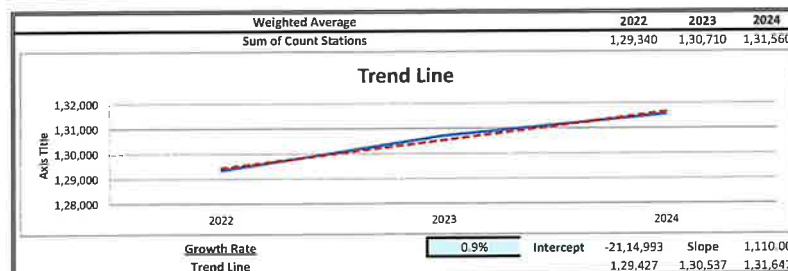
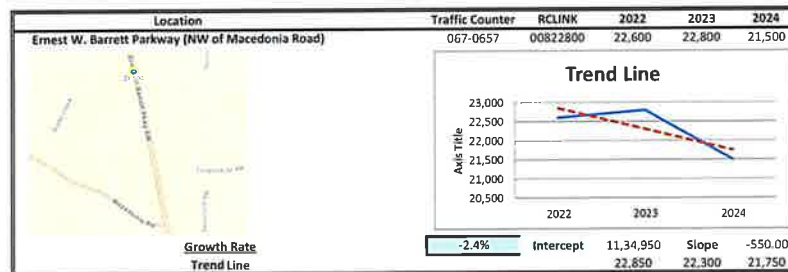
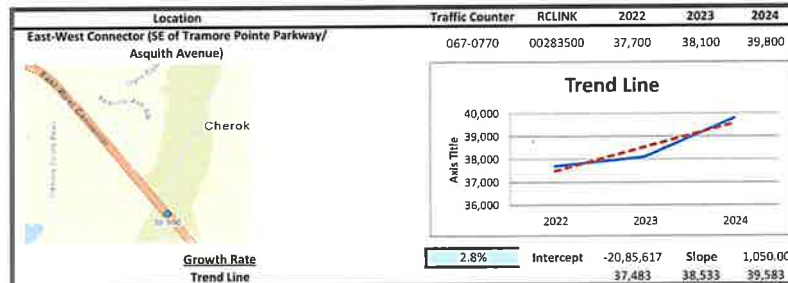
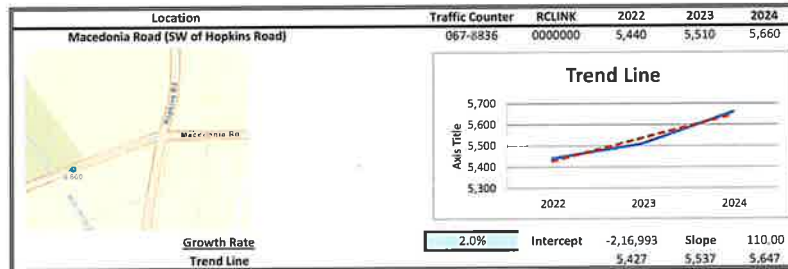
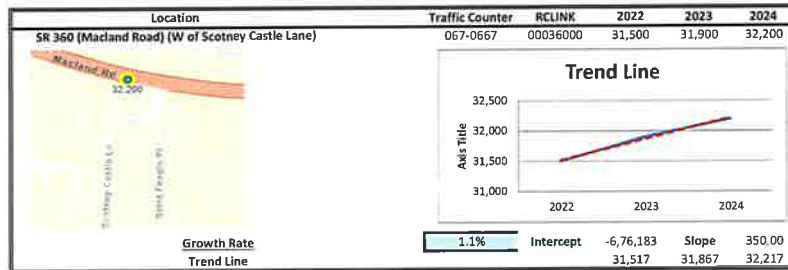
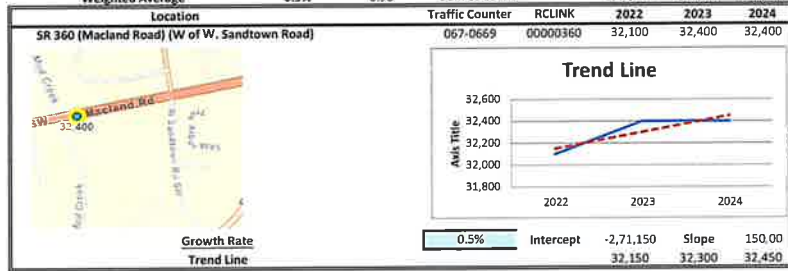
File Name : 20250375
Site Code : 20250375
Start Date : 10-22-2025
Page No : 3

	Ernest W Barrett Pkwy Northbound				Ernest W Barrett Pkwy Southbound				Macedonia Road Eastbound				Westbound				
Start Time	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Int. Total
Peak Hour Analysis From 04:00 PM to 05:45 PM - Peak 1 of 1																	
Peak Hour for Entire Intersection Begins at 05:00 PM																	
05:00 PM	107	307	0	414	0	243	14	257	2	0	62	64	0	0	0	0	735
05:15 PM	130	287	0	417	0	212	10	222	5	0	63	68	0	0	0	0	707
05:30 PM	116	290	0	406	0	248	19	267	3	0	55	58	0	0	0	0	731
05:45 PM	149	321	0	470	0	230	13	243	10	0	74	84	0	0	0	0	797
Total Volume	502	1205	0	1707	0	933	56	989	20	0	254	274	0	0	0	0	2970
% App. Total	29.4	70.6	0		0	94.3	5.7		7.3	0	92.7		0	0	0		
PHF	.842	.938	.000	.908	.000	.941	.737	.926	.500	.000	.858	.815	.000	.000	.000	.000	.932



LINEAR REGRESSION OF DAILY TRAFFIC

Location	Growth Rate	R Squared	Station ID	Route	2022	2023	2024
SR 360 (Macland Road) (W of W. Sandtown I	0.5%	0.75	067-0669	00000360	32,100	32,400	32,400
SR 360 (Macland Road) (W of Scotney Castle	1.1%	0.99	067-0667	00036000	31,500	31,900	32,200
Macedonia Road (SW of Hopkins Road)	2.0%	0.96	067-8836	00000000	5,440	5,510	5,660
East-West Connector (SE of Tramore Pointe I	2.8%	0.89	067-0770	00283500	37,700	38,100	39,800
Ernest W. Barrett Parkway (NW of Macedonia	-2.4%	0.62	067-0657	00822800	22,600	22,800	21,500
Weighted Average	0.9%	0.98	Sum of Count Stations =		1,29,340	1,30,710	1,31,560



EXISTING INTERSECTION ANALYSIS

Timings

1: Ernest W. Barrett Pkwy & Macedonia Rd

1a. Existing 2025 AM

11/18/2025



Lane Group	EBL	EBR	NBL	NBT	SBT
Lane Configurations					
Traffic Volume (vph)	66	511	154	722	940
Future Volume (vph)	66	511	154	722	940
Lane Group Flow (vph)	320	315	169	793	1059
Turn Type	Prot	Prot	pm+pt	NA	NA
Protected Phases	8	8	1	6	2
Permitted Phases			6		
Detector Phase	8	8	1	6	2
Switch Phase					
Minimum Initial (s)	6.0	6.0	5.0	15.0	15.0
Minimum Split (s)	23.5	23.5	15.0	23.5	40.5
Total Split (s)	40.0	40.0	30.0	80.0	50.0
Total Split (%)	33.3%	33.3%	25.0%	66.7%	41.7%
Yellow Time (s)	3.5	3.5	3.5	3.5	3.5
All-Red Time (s)	2.0	2.0	2.0	2.0	2.0
Lost Time Adjust (s)	0.0	0.0	0.0	0.0	0.0
Total Lost Time (s)	5.5	5.5	5.5	5.5	5.5
Lead/Lag			Lead		Lag
Lead-Lag Optimize?			Yes		Yes
Recall Mode	None	None	None	C-Min	C-Min
v/c Ratio	0.83	0.62	0.42	0.30	0.50
Control Delay (s/veh)	43.4	9.7	8.7	6.3	16.5
Queue Delay	0.0	0.0	0.0	0.0	0.0
Total Delay (s/veh)	43.4	9.7	8.7	6.3	16.5
Queue Length 50th (ft)	138	0	32	92	228
Queue Length 95th (ft)	222	76	74	166	392
Internal Link Dist (ft)	433			445	3140
Turn Bay Length (ft)	135		315		
Base Capacity (vph)	569	656	558	2629	2122
Starvation Cap Reductn	0	0	0	0	0
Spillback Cap Reductn	0	0	0	0	0
Storage Cap Reductn	0	0	0	0	0
Reduced v/c Ratio	0.56	0.48	0.30	0.30	0.50

Intersection Summary

Cycle Length: 120

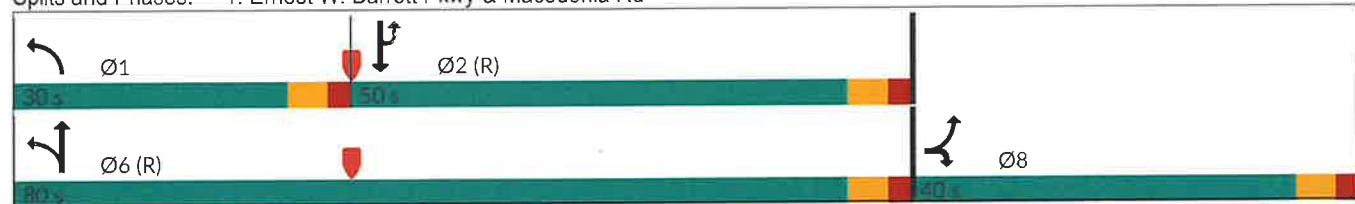
Actuated Cycle Length: 120

Offset: 0 (0%), Referenced to phase 2:SBTU and 6:NBTL, Start of Green

Natural Cycle: 80

Control Type: Actuated-Coordinated

Splits and Phases: 1: Ernest W. Barrett Pkwy & Macedonia Rd



HCM 6th Signalized Intersection Summary
1: Ernest W. Barrett Pkwy & Macedonia Rd

1a. Existing 2025 AM
11/18/2025

							
Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	66	511	154	722	0	940	24
Future Volume (veh/h)	66	511	154	722	0	940	24
Initial Q (Qb), veh	0	0	0	0		0	0
Ped-Bike Adj(A_pbT)	1.00	1.00	1.00				1.00
Parking Bus, Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	640	169	793		1033	26
Peak Hour Factor	0.91	0.91	0.91	0.91		0.91	0.91
Percent Heavy Veh, %	2	2	2	2		2	2
Cap, veh/h	404	719	376	2422		2055	52
Arrive On Green	0.00	0.23	0.06	0.68		0.58	0.58
Sat Flow, veh/h	1781	3170	1781	3647		3636	89
Grp Volume(v), veh/h	0	640	169	793		518	541
Grp Sat Flow(s),veh/h/ln	1781	1585	1781	1777		1777	1854
Q Serve(g_s), s	0.0	23.5	4.4	11.0		20.7	20.7
Cycle Q Clear(g_c), s	0.0	23.5	4.4	11.0		20.7	20.7
Prop In Lane	1.00	1.00	1.00				0.05
Lane Grp Cap(c), veh/h	404	719	376	2422		1031	1076
V/C Ratio(X)	0.00	0.89	0.45	0.33		0.50	0.50
Avail Cap(c_a), veh/h	512	911	641	2422		1031	1076
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Upstream Filter(I)	0.00	1.00	1.00	1.00		1.00	1.00
Uniform Delay (d), s/veh	0.0	45.0	11.2	7.8		14.9	14.9
Incr Delay (d2), s/veh	0.0	9.1	0.8	0.4		1.8	1.7
Initial Q Delay(d3), s/veh	0.0	0.0	0.0	0.0		0.0	0.0
%ile BackOfQ(50%),veh/ln	0.0	10.0	1.6	3.7		8.1	8.4
Unsig. Movement Delay, s/veh							
LnGrp Delay(d), s/veh	0.0	54.1	12.0	8.2		16.7	16.6
LnGrp LOS		D	B	A		B	B
Approach Vol, veh/h	640			962		1059	
Approach Delay, s/veh	54.1			8.9		16.7	
Approach LOS	D			A		B	
Timer - Assigned Phs	1	2				6	8
Phs Duration (G+Y+Rc), s	12.2	75.1				87.3	32.7
Change Period (Y+Rc), s	5.5	5.5				5.5	5.5
Max Green Setting (Gmax), s	24.5	44.5				74.5	34.5
Max Q Clear Time (g_c+l1), s	6.4	22.7				13.0	25.5
Green Ext Time (p_c), s	0.3	9.0				8.7	1.7
Intersection Summary							
HCM 6th Ctrl Delay, s/veh			22.8				
HCM 6th LOS			C				
Notes							
User approved volume balancing among the lanes for turning movement.							
User approved ignoring U-Turning movement.							

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

1a. Existing 2025 AM
11/18/2025

								
Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR	
Lane Configurations								
Traffic Volume (veh/h)	66	511	154	722	0	940	24	
Future Volume (veh/h)	66	511	154	722	0	940	24	
Number	3	18	1	6		2	12	
Initial Q, veh	0	0	0	0		0	0	
Ped-Bike Adj (A_pbT)	1.00	1.00	1.00				1.00	
Parking Bus Adj	1.00	1.00	1.00	1.00		1.00	1.00	
Work Zone On Approach	No			No		No		
Lanes Open During Work Zone								
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870	
Adj Flow Rate, veh/h	0	640	169	793		1033	26	
Peak Hour Factor	0.91	0.91	0.91	0.91		0.91	0.91	
Percent Heavy Veh, %	2	2	2	2		2	2	
Opposing Right Turn Influence	Yes		Yes					
Cap, veh/h	404	719	376	2422		2055	52	
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00	
Prop Arrive On Green	0.00	0.23	0.06	0.68		0.58	0.58	
Unsig. Movement Delay								
Ln Grp Delay, s/veh	0.0	54.1	12.0	8.2		16.7	16.6	
Ln Grp LOS		D	B	A		B	B	
Approach Vol, veh/h	640			962		1059		
Approach Delay, s/veh	54.1			8.9		16.7		
Approach LOS	D			A		B		
Timer:	1	2	3	4	5	6	7	8
Assigned Phs	1	2	8			6		
Case No	1.2	8.0	9.0			4.0		
Phs Duration (G+Y+Rc), s	12.2	75.1	32.7			87.3		
Change Period (Y+Rc), s	5.5	5.5	5.5			5.5		
Max Green (Gmax), s	24.5	44.5	34.5			74.5		
Max Allow Headway (MAH), s	3.5	5.8	3.8			5.8		
Max Q Clear (g_c+I1), s	6.4	22.7	25.5			13.0		
Green Ext Time (g_e), s	0.3	9.0	1.7			8.7		
Prob of Phs Call (p_c)	1.00	1.00	1.00			1.00		
Prob of Max Out (p_x)	0.00	0.30	0.12			0.00		
Left-Turn Movement Data								
Assigned Mvmt	1	5	3					
Mvmt Sat Flow, veh/h	1781	0	1781					
Through Movement Data								
Assigned Mvmt		2	8			6		
Mvmt Sat Flow, veh/h		3636	0			3647		
Right-Turn Movement Data								
Assigned Mvmt		12	18			16		
Mvmt Sat Flow, veh/h		89	3170			0		
Left Lane Group Data								
Assigned Mvmt	1	5	3	0	0	0	0	0
Lane Assignment	L (Pr/Pm)			L				

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

1a. Existing 2025 AM
11/18/2025

Lanes in Grp	1	0	1	0	0	0	0	0
Grp Vol (v), veh/h	169	0	0	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	1781	0	1781	0	0	0	0	0
Q Serve Time (g_s), s	4.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	4.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Sat Flow (s_l), veh/h/ln	533	0	1781	0	0	0	0	0
Shared LT Sat Flow (s_sh), veh/h/ln	0	0	0	0	0	0	0	0
Perm LT Eff Green (g_p), s	71.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Serve Time (g_u), s	48.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Q Serve Time (g_ps), s	10.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Time to First Blk (g_f), s	0.0	69.6	0.0	0.0	0.0	0.0	0.0	0.0
Serve Time pre Blk (g_fs), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop LT Inside Lane (P_L)	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	376	0	404	0	0	0	0	0
V/C Ratio (X)	0.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	641	0	512	0	0	0	0	0
Upstream Filter (I)	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	11.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	12.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
%ile Back of Q (50%), veh/ln	1.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Middle Lane Group Data								
Assigned Mvmt	0	2	8	0	0	6	0	0
Lane Assignment	T			T				
Lanes in Grp	0	1	0	0	0	2	0	0
Grp Vol (v), veh/h	0	518	0	0	0	793	0	0
Grp Sat Flow (s), veh/h/ln	0	1777	0	0	0	1777	0	0
Q Serve Time (g_s), s	0.0	20.7	0.0	0.0	0.0	11.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	20.7	0.0	0.0	0.0	11.0	0.0	0.0
Lane Grp Cap (c), veh/h	0	1031	0	0	0	2422	0	0
V/C Ratio (X)	0.00	0.50	0.00	0.00	0.00	0.33	0.00	0.00
Avail Cap (c_a), veh/h	0	1031	0	0	0	2422	0	0
Upstream Filter (I)	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	14.9	0.0	0.0	0.0	7.8	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.8	0.0	0.0	0.0	0.4	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	16.7	0.0	0.0	0.0	8.2	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	7.6	0.0	0.0	0.0	3.6	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.0	0.0	0.0	0.1	0.0	0.0

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

1a. Existing 2025 AM
11/18/2025

3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	8.1	0.0	0.0	0.0	3.7	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.07	0.00	0.00	0.00	0.19	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Right Lane Group Data

Assigned Mvmt	0	12	18	0	0	16	0	0
Lane Assignment		T+R	R					
Lanes in Grp	0	1	2	0	0	0	0	0
Grp Vol (v), veh/h	0	541	640	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	0	1854	1585	0	0	0	0	0
Q Serve Time (g_s), s	0.0	20.7	23.5	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	20.7	23.5	0.0	0.0	0.0	0.0	0.0
Prot RT Sat Flow (s_R), veh/h/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prot RT Eff Green (g_R), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop RT Outside Lane (P_R)	0.00	0.05	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	0	1076	719	0	0	0	0	0
V/C Ratio (X)	0.00	0.50	0.89	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	0	1076	911	0	0	0	0	0
Upstream Filter (I)	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	14.9	45.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.7	9.1	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	16.6	54.1	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	7.9	9.1	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.9	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	8.4	10.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.07	0.55	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Intersection Summary

HCM 6th Ctrl Delay, s/veh	22.8
HCM 6th LOS	C

Notes

User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

Timings

1: Ernest W. Barrett Pkwy & Macedonia Rd

1b. Existing 2025 PM

11/18/2025

	↖	↗	↙	↑	↓
Lane Group	EBL	EBR	NBL	NBT	SBT
Lane Configurations	↖	↗	↙	↑↑	↓↑
Traffic Volume (vph)	20	254	502	1205	933
Future Volume (vph)	20	254	502	1205	933
Lane Group Flow (vph)	148	147	540	1296	1063
Turn Type	Prot	Prot	pm+pt	NA	NA
Protected Phases	8	8	1	6	2
Permitted Phases			6		
Detector Phase	8	8	1	6	2
Switch Phase					
Minimum Initial (s)	6.0	6.0	5.0	15.0	15.0
Minimum Split (s)	23.5	23.5	15.0	23.5	40.5
Total Split (s)	30.0	30.0	45.0	120.0	75.0
Total Split (%)	20.0%	20.0%	30.0%	80.0%	50.0%
Yellow Time (s)	3.5	3.5	3.5	3.5	3.5
All-Red Time (s)	2.0	2.0	2.0	2.0	2.0
Lost Time Adjust (s)	0.0	0.0	0.0	0.0	0.0
Total Lost Time (s)	5.5	5.5	5.5	5.5	5.5
Lead/Lag			Lead		Lag
Lead-Lag Optimize?			Yes		Yes
Recall Mode	None	None	None	C-Min	C-Min
v/c Ratio	0.69	0.64	0.78	0.42	0.56
Control Delay (s/veh)	31.8	22.5	29.0	2.8	25.3
Queue Delay	0.0	0.0	0.0	0.0	0.0
Total Delay (s/veh)	31.8	22.5	29.0	2.8	25.3
Queue Length 50th (ft)	21	0	256	94	350
Queue Length 95th (ft)	92	73	420	176	490
Internal Link Dist (ft)	433			445	3140
Turn Bay Length (ft)	135		315		
Base Capacity (vph)	368	368	707	3064	1905
Starvation Cap Reductn	0	0	0	0	0
Spillback Cap Reductn	0	0	0	0	0
Storage Cap Reductn	0	0	0	0	0
Reduced v/c Ratio	0.40	0.40	0.76	0.42	0.56

Intersection Summary

Cycle Length: 150

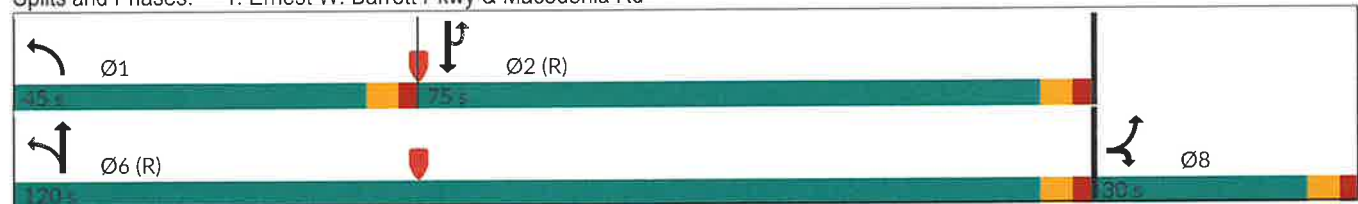
Actuated Cycle Length: 150

Offset: 0 (0%), Referenced to phase 2:SBTU and 6:NBTL, Start of Green

Natural Cycle: 90

Control Type: Actuated-Coordinated

Splits and Phases: 1: Ernest W. Barrett Pkwy & Macedonia Rd



HCM 6th Signalized Intersection Summary
1: Ernest W. Barrett Pkwy & Macedonia Rd

1b. Existing 2025 PM
11/18/2025

							
Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	20	254	502	1205	0	933	56
Future Volume (veh/h)	20	254	502	1205	0	933	56
Initial Q (Qb), veh	0	0	0	0		0	0
Ped-Bike Adj(A_pbT)	1.00	1.00	1.00				1.00
Parking Bus, Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	297	540	1296		1003	60
Peak Hour Factor	0.93	0.93	0.93	0.93		0.93	0.93
Percent Heavy Veh, %	2	2	2	2		2	2
Cap, veh/h	196	348	568	2903		2145	128
Arrive On Green	0.00	0.11	0.15	0.82		0.63	0.63
Sat Flow, veh/h	1781	3170	1781	3647		3500	204
Grp Volume(v), veh/h	0	297	540	1296		523	540
Grp Sat Flow(s),veh/h/ln	1781	1585	1781	1777		1777	1834
Q Serve(g_s), s	0.0	13.8	19.2	15.8		23.2	23.2
Cycle Q Clear(g_c), s	0.0	13.8	19.2	15.8		23.2	23.2
Prop In Lane	1.00	1.00	1.00				0.11
Lane Grp Cap(c), veh/h	196	348	568	2903		1119	1155
V/C Ratio(X)	0.00	0.85	0.95	0.45		0.47	0.47
Avail Cap(c_a), veh/h	291	518	769	2903		1119	1155
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Upstream Filter(I)	0.00	1.00	1.00	1.00		1.00	1.00
Uniform Delay (d), s/veh	0.0	65.6	23.1	4.0		14.6	14.6
Incr Delay (d2), s/veh	0.0	8.8	17.9	0.5		1.4	1.4
Initial Q Delay(d3), s/veh	0.0	0.0	0.0	0.0		0.0	0.0
%ile BackOfQ(50%),veh/ln	0.0	6.0	22.1	4.3		9.2	9.5
Unsig. Movement Delay, s/veh							
LnGrp Delay(d), s/veh	0.0	74.4	41.0	4.5		16.0	15.9
LnGrp LOS		E	D	A		B	B
Approach Vol, veh/h	297			1836		1063	
Approach Delay, s/veh	74.4			15.2		16.0	
Approach LOS	E			B		B	
Timer - Assigned Phs	1	2				6	8
Phs Duration (G+Y+Rc), s	28.1	100.0				128.0	22.0
Change Period (Y+Rc), s	5.5	5.5				5.5	5.5
Max Green Setting (Gmax), s	39.5	69.5				114.5	24.5
Max Q Clear Time (g_c+l1), s	21.2	25.2				17.8	15.8
Green Ext Time (p_c), s	1.4	11.6				20.0	0.7
Intersection Summary							
HCM 6th Ctrl Delay, s/veh			21.0				
HCM 6th LOS			C				
Notes							
User approved pedestrian interval to be less than phase max green.							
User approved volume balancing among the lanes for turning movement.							
User approved ignoring U-Turning movement.							

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

1b. Existing 2025 PM
11/18/2025

								
Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR	
Lane Configurations								
Traffic Volume (veh/h)	20	254	502	1205	0	933	56	
Future Volume (veh/h)	20	254	502	1205	0	933	56	
Number	3	18	1	6		2	12	
Initial Q, veh	0	0	0	0		0	0	
Ped-Bike Adj (A_pbT)	1.00	1.00	1.00				1.00	
Parking Bus Adj	1.00	1.00	1.00	1.00		1.00	1.00	
Work Zone On Approach	No			No		No		
Lanes Open During Work Zone								
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870	
Adj Flow Rate, veh/h	0	297	540	1296		1003	60	
Peak Hour Factor	0.93	0.93	0.93	0.93		0.93	0.93	
Percent Heavy Veh, %	2	2	2	2		2	2	
Opposing Right Turn Influence	Yes		Yes					
Cap, veh/h	196	348	568	2903		2145	128	
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00	
Prop Arrive On Green	0.00	0.11	0.15	0.82		0.63	0.63	
Unsig. Movement Delay								
Ln Grp Delay, s/veh	0.0	74.4	41.0	4.5		16.0	15.9	
Ln Grp LOS		E	D	A		B	B	
Approach Vol, veh/h	297			1836		1063		
Approach Delay, s/veh	74.4			15.2		16.0		
Approach LOS	E			B		B		
Timer:	1	2	3	4	5	6	7	8
Assigned Phs	1	2	8			6		
Case No	1.2	8.0	9.0			4.0		
Phs Duration (G+Y+Rc), s	28.1	100.0	22.0			128.0		
Change Period (Y+Rc), s	5.5	5.5	5.5			5.5		
Max Green (Gmax), s	39.5	69.5	24.5			114.5		
Max Allow Headway (MAH), s	3.5	5.8	3.8			5.8		
Max Q Clear (g_c+I1), s	21.2	25.2	15.8			17.8		
Green Ext Time (g_e), s	1.4	11.6	0.7			20.0		
Prob of Phs Call (p_c)	1.00	1.00	1.00			1.00		
Prob of Max Out (p_x)	0.00	0.06	0.03			0.01		
Left-Turn Movement Data								
Assigned Mvmt	1	5	3					
Mvmt Sat Flow, veh/h	1781	0	1781					
Through Movement Data								
Assigned Mvmt		2	8			6		
Mvmt Sat Flow, veh/h		3500	0			3647		
Right-Turn Movement Data								
Assigned Mvmt		12	18			16		
Mvmt Sat Flow, veh/h		204	3170			0		
Left Lane Group Data								
Assigned Mvmt	1	5	3	0	0	0	0	0
Lane Assignment	L (Pr/Pm)			L				

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

1b. Existing 2025 PM
11/18/2025

Lanes in Grp	1	0	1	0	0	0	0	0
Grp Vol (v), veh/h	540	0	0	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	1781	0	1781	0	0	0	0	0
Q Serve Time (g_s), s	19.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	19.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Sat Flow (s_l), veh/h/ln	531	0	1781	0	0	0	0	0
Shared LT Sat Flow (s_sh), veh/h/ln	0	0	0	0	0	0	0	0
Perm LT Eff Green (g_p), s	96.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Serve Time (g_u), s	71.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Q Serve Time (g_ps), s	71.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Time to First Blk (g_f), s	0.0	94.5	0.0	0.0	0.0	0.0	0.0	0.0
Serve Time pre Blk (g_fs), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop LT Inside Lane (P_L)	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	568	0	196	0	0	0	0	0
V/C Ratio (X)	0.95	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	769	0	291	0	0	0	0	0
Upstream Filter (I)	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	23.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	17.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	41.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	19.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	2.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
%ile Back of Q (50%), veh/ln	22.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	1.78	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Middle Lane Group Data								
Assigned Mvmt	0	2	8	0	0	6	0	0
Lane Assignment	T				T			
Lanes in Grp	0	1	0	0	0	2	0	0
Grp Vol (v), veh/h	0	523	0	0	0	1296	0	0
Grp Sat Flow (s), veh/h/ln	0	1777	0	0	0	1777	0	0
Q Serve Time (g_s), s	0.0	23.2	0.0	0.0	0.0	15.8	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	23.2	0.0	0.0	0.0	15.8	0.0	0.0
Lane Grp Cap (c), veh/h	0	1119	0	0	0	2903	0	0
V/C Ratio (X)	0.00	0.47	0.00	0.00	0.00	0.45	0.00	0.00
Avail Cap (c_a), veh/h	0	1119	0	0	0	2903	0	0
Upstream Filter (I)	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	14.6	0.0	0.0	0.0	4.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.4	0.0	0.0	0.0	0.5	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	16.0	0.0	0.0	0.0	4.5	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	8.8	0.0	0.0	0.0	4.1	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.4	0.0	0.0	0.0	0.2	0.0	0.0

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

1b. Existing 2025 PM
11/18/2025

3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	9.2	0.0	0.0	0.0	4.3	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.07	0.00	0.00	0.00	0.23	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Right Lane Group Data

Assigned Mvmt	0	12	18	0	0	16	0	0
Lane Assignment		T+R	R					
Lanes in Grp	0	1	2	0	0	0	0	0
Grp Vol (v), veh/h	0	540	297	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	0	1834	1585	0	0	0	0	0
Q Serve Time (g_s), s	0.0	23.2	13.8	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	23.2	13.8	0.0	0.0	0.0	0.0	0.0
Prot RT Sat Flow (s_R), veh/h/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prot RT Eff Green (g_R), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop RT Outside Lane (P_R)	0.00	0.11	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	0	1155	348	0	0	0	0	0
V/C Ratio (X)	0.00	0.47	0.85	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	0	1155	518	0	0	0	0	0
Upstream Filter (I)	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	14.6	65.6	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.4	8.8	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	15.9	74.4	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	9.1	5.6	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.4	0.4	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	9.5	6.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.08	0.33	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Intersection Summary

HCM 6th Ctrl Delay, s/veh	21.0
HCM 6th LOS	C

Notes

User approved pedestrian interval to be less than phase max green.
User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

**FUTURE "NO-BUILD" INTERSECTION
ANALYSIS**

Timings

1: Ernest W. Barrett Pkwy & Macedonia Rd

2a. No Build 2027 AM

11/18/2025



Lane Group	EBL	EBR	NBL	NBT	SBT
Lane Configurations					
Traffic Volume (vph)	67	521	157	736	959
Future Volume (vph)	67	521	157	736	959
Lane Group Flow (vph)	326	321	173	809	1080
Turn Type	Prot	Prot	pm+pt	NA	NA
Protected Phases	8	8	1	6	2
Permitted Phases			6		
Detector Phase	8	8	1	6	2
Switch Phase					
Minimum Initial (s)	6.0	6.0	5.0	15.0	15.0
Minimum Split (s)	23.5	23.5	15.0	23.5	40.5
Total Split (s)	40.0	40.0	30.0	80.0	50.0
Total Split (%)	33.3%	33.3%	25.0%	66.7%	41.7%
Yellow Time (s)	3.5	3.5	3.5	3.5	3.5
All-Red Time (s)	2.0	2.0	2.0	2.0	2.0
Lost Time Adjust (s)	0.0	0.0	0.0	0.0	0.0
Total Lost Time (s)	5.5	5.5	5.5	5.5	5.5
Lead/Lag			Lead		Lag
Lead-Lag Optimize?			Yes		Yes
Recall Mode	None	None	None	C-Min	C-Min
v/c Ratio	0.83	0.62	0.43	0.31	0.52
Control Delay (s/veh)	43.6	9.6	9.1	6.4	17.4
Queue Delay	0.0	0.0	0.0	0.0	0.0
Total Delay (s/veh)	43.6	9.6	9.1	6.4	17.4
Queue Length 50th (ft)	142	0	34	96	241
Queue Length 95th (ft)	227	76	77	172	411
Internal Link Dist (ft)	433			445	3140
Turn Bay Length (ft)	135		315		
Base Capacity (vph)	570	661	547	2617	2093
Starvation Cap Reductn	0	0	0	0	0
Spillback Cap Reductn	0	0	0	0	0
Storage Cap Reductn	0	0	0	0	0
Reduced v/c Ratio	0.57	0.49	0.32	0.31	0.52

Intersection Summary

Cycle Length: 120

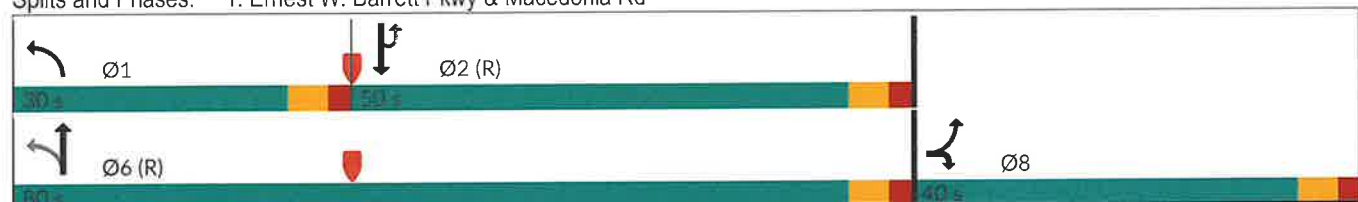
Actuated Cycle Length: 120

Offset: 0 (0%), Referenced to phase 2:SBTU and 6:NBTL, Start of Green

Natural Cycle: 80

Control Type: Actuated-Coordinated

Splits and Phases: 1: Ernest W. Barrett Pkwy & Macedonia Rd



A&R Engineering, Inc

25-196 Res Dev at Old Horseshoe Bend Rd and Ernest Barrett Pkwy - Cobb County, GA

Synchro 11 Light Report

Page 1

HCM 6th Signalized Intersection Summary
1: Ernest W. Barrett Pkwy & Macedonia Rd

2a. No Build 2027 AM
11/18/2025

							
Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	67	521	157	736	0	959	24
Future Volume (veh/h)	67	521	157	736	0	959	24
Initial Q (Qb), veh	0	0	0	0		0	0
Ped-Bike Adj(A_pbT)	1.00	1.00	1.00				1.00
Parking Bus, Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	652	173	809		1054	26
Peak Hour Factor	0.91	0.91	0.91	0.91		0.91	0.91
Percent Heavy Veh, %	2	2	2	2		2	2
Cap, veh/h	410	730	368	2409		2038	50
Arrive On Green	0.00	0.23	0.06	0.68		0.58	0.58
Sat Flow, veh/h	1781	3170	1781	3647		3638	87
Grp Volume(v), veh/h	0	652	173	809		528	552
Grp Sat Flow(s),veh/h/ln	1781	1585	1781	1777		1777	1855
Q Serve(g_s), s	0.0	23.9	4.5	11.4		21.6	21.6
Cycle Q Clear(g_c), s	0.0	23.9	4.5	11.4		21.6	21.6
Prop In Lane	1.00	1.00	1.00				0.05
Lane Grp Cap(c), veh/h	410	730	368	2409		1022	1066
V/C Ratio(X)	0.00	0.89	0.47	0.34		0.52	0.52
Avail Cap(c_a), veh/h	512	911	630	2409		1022	1066
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Upstream Filter(I)	0.00	1.00	1.00	1.00		1.00	1.00
Uniform Delay (d), s/veh	0.0	44.7	11.7	8.1		15.4	15.4
Incr Delay (d2), s/veh	0.0	9.5	0.9	0.4		1.9	1.8
Initial Q Delay(d3), s/veh	0.0	0.0	0.0	0.0		0.0	0.0
%ile BackOfQ(50%),veh/ln	0.0	10.2	1.6	3.9		8.5	8.8
Unsig. Movement Delay, s/veh							
LnGrp Delay(d), s/veh	0.0	54.2	12.6	8.4		17.3	17.2
LnGrp LOS		D	B	A		B	B
Approach Vol, veh/h	652			982		1080	
Approach Delay, s/veh	54.2			9.2		17.3	
Approach LOS	D			A		B	
Timer - Assigned Phs	1	2				6	8
Phs Duration (G+Y+Rc), s	12.4	74.5				86.9	33.1
Change Period (Y+Rc), s	5.5	5.5				5.5	5.5
Max Green Setting (Gmax), s	24.5	44.5				74.5	34.5
Max Q Clear Time (g_c+l1), s	6.5	23.6				13.4	25.9
Green Ext Time (p_c), s	0.3	9.0				8.9	1.7
Intersection Summary							
HCM 6th Ctrl Delay, s/veh			23.2				
HCM 6th LOS			C				
Notes							
User approved volume balancing among the lanes for turning movement.							
User approved ignoring U-Turning movement.							

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

2a. No Build 2027 AM
11/18/2025

									
Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR		
Lane Configurations									
Traffic Volume (veh/h)	67	521	157	736	0	959	24		
Future Volume (veh/h)	67	521	157	736	0	959	24		
Number	3	18	1	6		2	12		
Initial Q, veh	0	0	0	0		0	0		
Ped-Bike Adj (A _{pbT})	1.00	1.00	1.00				1.00		
Parking Bus Adj	1.00	1.00	1.00	1.00		1.00	1.00		
Work Zone On Approach	No			No		No			
Lanes Open During Work Zone									
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870		
Adj Flow Rate, veh/h	0	652	173	809		1054	26		
Peak Hour Factor	0.91	0.91	0.91	0.91		0.91	0.91		
Percent Heavy Veh, %	2	2	2	2		2	2		
Opposing Right Turn Influence	Yes		Yes						
Cap, veh/h	410	730	368	2409		2038	50		
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00		
Prop Arrive On Green	0.00	0.23	0.06	0.68		0.58	0.58		
Unsig. Movement Delay									
Ln Grp Delay, s/veh	0.0	54.2	12.6	8.4		17.3	17.2		
Ln Grp LOS		D	B	A		B	B		
Approach Vol, veh/h	652			982		1080			
Approach Delay, s/veh	54.2			9.2		17.3			
Approach LOS	D			A		B			
Timer:		1	2	3	4	5	6	7	8
Assigned Phs		1	2	8			6		
Case No		1.2	8.0	9.0			4.0		
Phs Duration (G+Y+Rc), s		12.4	74.5	33.1			86.9		
Change Period (Y+Rc), s		5.5	5.5	5.5			5.5		
Max Green (Gmax), s		24.5	44.5	34.5			74.5		
Max Allow Headway (MAH), s		3.5	5.8	3.8			5.8		
Max Q Clear (g _c +l1), s		6.5	23.6	25.9			13.4		
Green Ext Time (g _e), s		0.3	9.0	1.7			8.9		
Prob of Phs Call (p _c)		1.00	1.00	1.00			1.00		
Prob of Max Out (p _x)		0.00	0.33	0.15			0.00		
Left-Turn Movement Data									
Assigned Mvmt		1	5	3					
Mvmt Sat Flow, veh/h		1781	0	1781					
Through Movement Data									
Assigned Mvmt			2	8			6		
Mvmt Sat Flow, veh/h			3638	0			3647		
Right-Turn Movement Data									
Assigned Mvmt			12	18			16		
Mvmt Sat Flow, veh/h			87	3170			0		
Left Lane Group Data									
Assigned Mvmt		1	5	3	0	0	0	0	0
Lane Assignment		L (Pr/Pm)		L					

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

2a. No Build 2027 AM
11/18/2025

Lanes in Grp	1	0	1	0	0	0	0	0
Grp Vol (v), veh/h	173	0	0	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	1781	0	1781	0	0	0	0	0
Q Serve Time (g_s), s	4.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	4.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Sat Flow (s_l), veh/h/ln	522	0	1781	0	0	0	0	0
Shared LT Sat Flow (s_sh), veh/h/ln	0	0	0	0	0	0	0	0
Perm LT Eff Green (g_p), s	71.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Serve Time (g_u), s	47.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Q Serve Time (g_ps), s	11.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Time to First Blk (g_f), s	0.0	69.0	0.0	0.0	0.0	0.0	0.0	0.0
Serve Time pre Blk (g_fs), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop LT Inside Lane (P_L)	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	368	0	410	0	0	0	0	0
V/C Ratio (X)	0.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	630	0	512	0	0	0	0	0
Upstream Filter (I)	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	11.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	12.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	1.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
%ile Back of Q (50%), veh/ln	1.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Middle Lane Group Data								
Assigned Mvmt	0	2	8	0	0	6	0	0
Lane Assignment	T				T			
Lanes in Grp	0	1	0	0	0	2	0	0
Grp Vol (v), veh/h	0	528	0	0	0	809	0	0
Grp Sat Flow (s), veh/h/ln	0	1777	0	0	0	1777	0	0
Q Serve Time (g_s), s	0.0	21.6	0.0	0.0	0.0	11.4	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	21.6	0.0	0.0	0.0	11.4	0.0	0.0
Lane Grp Cap (c), veh/h	0	1022	0	0	0	2409	0	0
V/C Ratio (X)	0.00	0.52	0.00	0.00	0.00	0.34	0.00	0.00
Avail Cap (c_a), veh/h	0	1022	0	0	0	2409	0	0
Upstream Filter (I)	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	15.4	0.0	0.0	0.0	8.1	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.9	0.0	0.0	0.0	0.4	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	17.3	0.0	0.0	0.0	8.4	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	8.0	0.0	0.0	0.0	3.8	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.0	0.0	0.0	0.1	0.0	0.0

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

2a. No Build 2027 AM
11/18/2025

3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	8.5	0.0	0.0	0.0	3.9	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.07	0.00	0.00	0.00	0.20	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Right Lane Group Data

Assigned Mvmt	0	12	18	0	0	16	0	0
Lane Assignment		T+R	R					
Lanes in Grp	0	1	2	0	0	0	0	0
Grp Vol (v), veh/h	0	552	652	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	0	1855	1585	0	0	0	0	0
Q Serve Time (g_s), s	0.0	21.6	23.9	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	21.6	23.9	0.0	0.0	0.0	0.0	0.0
Prot RT Sat Flow (s_R), veh/h/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prot RT Eff Green (g_R), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop RT Outside Lane (P_R)	0.00	0.05	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	0	1066	730	0	0	0	0	0
V/C Ratio (X)	0.00	0.52	0.89	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	0	1066	911	0	0	0	0	0
Upstream Filter (I)	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	15.4	44.7	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.8	9.5	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	17.2	54.2	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	8.3	9.3	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	1.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	8.8	10.2	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.07	0.56	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Intersection Summary

HCM 6th Ctrl Delay, s/veh	23.2
HCM 6th LOS	C

Notes

User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

Timings

1: Ernest W. Barrett Pkwy & Macedonia Rd

2b. No Build 2027 PM

11/18/2025

	EBL	EBR	NBL	NBT	SBT
Lane Group	EBL	EBR	NBL	NBT	SBT
Lane Configurations					
Traffic Volume (vph)	20	259	512	1229	952
Future Volume (vph)	20	259	512	1229	952
Lane Group Flow (vph)	150	150	551	1322	1085
Turn Type	Prot	Prot	pm+pt	NA	NA
Protected Phases	8	8	1	6	2
Permitted Phases			6		
Detector Phase	8	8	1	6	2
Switch Phase					
Minimum Initial (s)	6.0	6.0	5.0	15.0	15.0
Minimum Split (s)	23.5	23.5	15.0	23.5	40.5
Total Split (s)	30.0	30.0	45.0	120.0	75.0
Total Split (%)	20.0%	20.0%	30.0%	80.0%	50.0%
Yellow Time (s)	3.5	3.5	3.5	3.5	3.5
All-Red Time (s)	2.0	2.0	2.0	2.0	2.0
Lost Time Adjust (s)	0.0	0.0	0.0	0.0	0.0
Total Lost Time (s)	5.5	5.5	5.5	5.5	5.5
Lead/Lag			Lead		Lag
Lead-Lag Optimize?			Yes		Yes
Recall Mode	None	None	None	C-Min	C-Min
v/c Ratio	0.69	0.65	0.79	0.43	0.58
Control Delay (s/veh)	31.7	22.4	31.6	2.8	26.7
Queue Delay	0.0	0.0	0.0	0.0	0.0
Total Delay (s/veh)	31.7	22.4	31.6	2.8	26.7
Queue Length 50th (ft)	21	0	282	97	372
Queue Length 95th (ft)	94	73	454	183	504
Internal Link Dist (ft)	433			445	3140
Turn Bay Length (ft)	135		315		
Base Capacity (vph)	370	371	707	3064	1865
Starvation Cap Reductn	0	0	0	0	0
Spillback Cap Reductn	0	0	0	0	0
Storage Cap Reductn	0	0	0	0	0
Reduced v/c Ratio	0.41	0.40	0.78	0.43	0.58

Intersection Summary

Cycle Length: 150

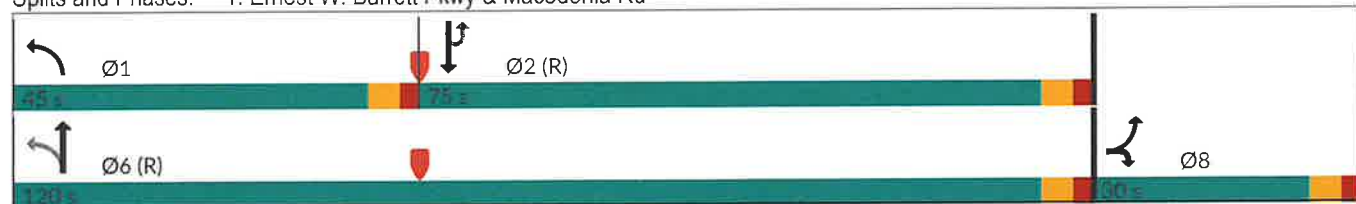
Actuated Cycle Length: 150

Offset: 0 (0%), Referenced to phase 2:SBTU and 6:NBTL, Start of Green

Natural Cycle: 90

Control Type: Actuated-Coordinated

Splits and Phases: 1: Ernest W. Barrett Pkwy & Macedonia Rd



A&R Engineering, Inc

25-196 Res Dev at Old Horseshoe Bend Rd and Ernest Barrett Pkwy - Cobb County, GA

Synchro 11 Light Report

Page 1

HCM 6th Signalized Intersection Summary
1: Ernest W. Barrett Pkwy & Macedonia Rd

2b. No Build 2027 PM
11/18/2025

							
Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	20	259	512	1229	0	952	57
Future Volume (veh/h)	20	259	512	1229	0	952	57
Initial Q (Qb), veh	0	0	0	0		0	0
Ped-Bike Adj(A_pbT)	1.00	1.00	1.00				1.00
Parking Bus, Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	302	551	1322		1024	61
Peak Hour Factor	0.93	0.93	0.93	0.93		0.93	0.93
Percent Heavy Veh, %	2	2	2	2		2	2
Cap, veh/h	198	353	579	2897		2077	124
Arrive On Green	0.00	0.11	0.17	0.82		0.61	0.61
Sat Flow, veh/h	1781	3170	1781	3647		3501	203
Grp Volume(v), veh/h	0	302	551	1322		534	551
Grp Sat Flow(s),veh/h/ln	1781	1585	1781	1777		1777	1834
Q Serve(g_s), s	0.0	14.0	22.0	16.4		25.2	25.2
Cycle Q Clear(g_c), s	0.0	14.0	22.0	16.4		25.2	25.2
Prop In Lane	1.00	1.00	1.00				0.11
Lane Grp Cap(c), veh/h	198	353	579	2897		1083	1118
V/C Ratio(X)	0.00	0.86	0.95	0.46		0.49	0.49
Avail Cap(c_a), veh/h	291	518	747	2897		1083	1118
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Upstream Filter(I)	0.00	1.00	1.00	1.00		1.00	1.00
Uniform Delay (d), s/veh	0.0	65.5	25.8	4.1		16.3	16.3
Incr Delay (d2), s/veh	0.0	9.1	19.2	0.5		1.6	1.6
Initial Q Delay(d3), s/veh	0.0	0.0	0.0	0.0		0.0	0.0
%ile BackOfQ(50%),veh/ln	0.0	6.1	22.7	4.6		10.2	10.5
Unsig. Movement Delay, s/veh							
LnGrp Delay(d), s/veh	0.0	74.6	45.1	4.6		17.9	17.9
LnGrp LOS		E	D	A		B	B
Approach Vol, veh/h	302			1873		1085	
Approach Delay, s/veh	74.6			16.5		17.9	
Approach LOS	E			B		B	
Timer - Assigned Phs	1	2				6	8
Phs Duration (G+Y+Rc), s	30.9	96.9				127.8	22.2
Change Period (Y+Rc), s	5.5	5.5				5.5	5.5
Max Green Setting (Gmax), s	39.5	69.5				114.5	24.5
Max Q Clear Time (g_c+I1), s	24.0	27.2				18.4	16.0
Green Ext Time (p_c), s	1.4	11.8				20.8	0.7
Intersection Summary							
HCM 6th Ctrl Delay, s/veh			22.4				
HCM 6th LOS			C				
Notes							
User approved pedestrian interval to be less than phase max green.							
User approved volume balancing among the lanes for turning movement.							
User approved ignoring U-Turning movement.							

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

2b. No Build 2027 PM
11/18/2025

									
Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR		
Lane Configurations									
Traffic Volume (veh/h)	20	259	512	1229	0	952	57		
Future Volume (veh/h)	20	259	512	1229	0	952	57		
Number	3	18	1	6		2	12		
Initial Q, veh	0	0	0	0		0	0		
Ped-Bike Adj (A_pbT)	1.00	1.00	1.00				1.00		
Parking Bus Adj	1.00	1.00	1.00	1.00		1.00	1.00		
Work Zone On Approach	No			No		No			
Lanes Open During Work Zone									
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870		
Adj Flow Rate, veh/h	0	302	551	1322		1024	61		
Peak Hour Factor	0.93	0.93	0.93	0.93		0.93	0.93		
Percent Heavy Veh, %	2	2	2	2		2	2		
Opposing Right Turn Influence	Yes		Yes						
Cap, veh/h	198	353	579	2897		2077	124		
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00		
Prop Arrive On Green	0.00	0.11	0.17	0.82		0.61	0.61		
Unsig. Movement Delay									
Ln Grp Delay, s/veh	0.0	74.6	45.1	4.6		17.9	17.9		
Ln Grp LOS		E	D	A		B	B		
Approach Vol, veh/h	302			1873		1085			
Approach Delay, s/veh	74.6			16.5		17.9			
Approach LOS	E			B		B			
Timer:		1	2	3	4	5	6	7	8
Assigned Phs		1	2	8			6		
Case No		1.2	8.0	9.0			4.0		
Phs Duration (G+Y+Rc), s		30.9	96.9	22.2			127.8		
Change Period (Y+Rc), s		5.5	5.5	5.5			5.5		
Max Green (Gmax), s		39.5	69.5	24.5			114.5		
Max Allow Headway (MAH), s		3.5	5.8	3.8			5.8		
Max Q Clear (g_c+I1), s		24.0	27.2	16.0			18.4		
Green Ext Time (g_e), s		1.4	11.8	0.7			20.8		
Prob of Phs Call (p_c)		1.00	1.00	1.00			1.00		
Prob of Max Out (p_x)		0.00	0.07	0.04			0.01		
Left-Turn Movement Data									
Assigned Mvmt		1	5	3					
Mvmt Sat Flow, veh/h		1781	0	1781					
Through Movement Data									
Assigned Mvmt			2	8			6		
Mvmt Sat Flow, veh/h			3501	0			3647		
Right-Turn Movement Data									
Assigned Mvmt			12	18			16		
Mvmt Sat Flow, veh/h			203	3170			0		
Left Lane Group Data									
Assigned Mvmt		1	5	3	0	0	0	0	0
Lane Assignment		L (Pr/Pm)			L				

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

2b. No Build 2027 PM
11/18/2025

Lanes in Grp	1	0	1	0	0	0	0	0
Grp Vol (v), veh/h	551	0	0	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	1781	0	1781	0	0	0	0	0
Q Serve Time (g_s), s	22.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	22.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Sat Flow (s_l), veh/h/ln	520	0	1781	0	0	0	0	0
Shared LT Sat Flow (s_sh), veh/h/ln	0	0	0	0	0	0	0	0
Perm LT Eff Green (g_p), s	93.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Serve Time (g_u), s	66.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Q Serve Time (g_ps), s	66.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Time to First Blk (g_f), s	0.0	91.4	0.0	0.0	0.0	0.0	0.0	0.0
Serve Time pre Blk (g_fs), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop LT Inside Lane (P_L)	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	579	0	198	0	0	0	0	0
V/C Ratio (X)	0.95	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	747	0	291	0	0	0	0	0
Upstream Filter (I)	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	25.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	19.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	45.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	19.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	3.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
%ile Back of Q (50%), veh/ln	22.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	1.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Middle Lane Group Data

Assigned Mvmt	0	2	8	0	0	6	0	0
Lane Assignment	T				T			
Lanes in Grp	0	1	0	0	0	2	0	0
Grp Vol (v), veh/h	0	534	0	0	0	1322	0	0
Grp Sat Flow (s), veh/h/ln	0	1777	0	0	0	1777	0	0
Q Serve Time (g_s), s	0.0	25.2	0.0	0.0	0.0	16.4	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	25.2	0.0	0.0	0.0	16.4	0.0	0.0
Lane Grp Cap (c), veh/h	0	1083	0	0	0	2897	0	0
V/C Ratio (X)	0.00	0.49	0.00	0.00	0.00	0.46	0.00	0.00
Avail Cap (c_a), veh/h	0	1083	0	0	0	2897	0	0
Upstream Filter (I)	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	16.3	0.0	0.0	0.0	4.1	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.6	0.0	0.0	0.0	0.5	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	17.9	0.0	0.0	0.0	4.6	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	9.7	0.0	0.0	0.0	4.4	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.0	0.0	0.0	0.2	0.0	0.0

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

2b. No Build 2027 PM
11/18/2025

3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	10.2	0.0	0.0	0.0	4.6	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.08	0.00	0.00	0.00	0.24	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Right Lane Group Data

Assigned Mvmt	0	12	18	0	0	16	0	0
Lane Assignment	T+R		R					
Lanes in Grp	0	1	2	0	0	0	0	0
Grp Vol (v), veh/h	0	551	302	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	0	1834	1585	0	0	0	0	0
Q Serve Time (g_s), s	0.0	25.2	14.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	25.2	14.0	0.0	0.0	0.0	0.0	0.0
Prot RT Sat Flow (s_R), veh/h/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prot RT Eff Green (g_R), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop RT Outside Lane (P_R)	0.00	0.11	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	0	1118	353	0	0	0	0	0
V/C Ratio (X)	0.00	0.49	0.86	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	0	1118	518	0	0	0	0	0
Upstream Filter (I)	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	16.3	65.5	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.6	9.1	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	17.9	74.6	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	10.0	5.7	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.4	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	10.5	6.1	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.08	0.34	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Intersection Summary

HCM 6th Ctrl Delay, s/veh	22.4
HCM 6th LOS	C

Notes

User approved pedestrian interval to be less than phase max green.
User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

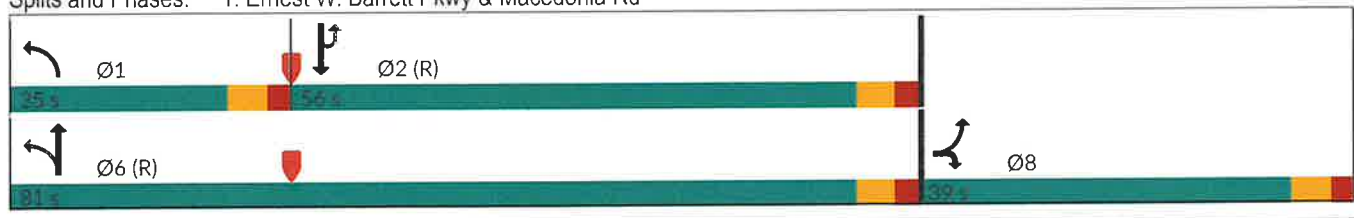
FUTURE "BUILD" INTERSECTION ANALYSIS

						
Lane Group	EBL	EBR	NBL	NBT	SBU	SBT
Lane Configurations						
Traffic Volume (vph)	68	521	157	741	3	972
Future Volume (vph)	68	521	157	741	3	972
Lane Group Flow (vph)	327	321	173	814	3	1098
Turn Type	Prot	Prot	pm+pt	NA	Perm	NA
Protected Phases	8	8	1	6		2
Permitted Phases			6		2	
Detector Phase	8	8	1	6	2	2
Switch Phase						
Minimum Initial (s)	6.0	6.0	5.0	15.0	15.0	15.0
Minimum Split (s)	23.5	23.5	15.0	23.5	40.5	40.5
Total Split (s)	39.0	39.0	25.0	81.0	56.0	56.0
Total Split (%)	32.5%	32.5%	20.8%	67.5%	46.7%	46.7%
Yellow Time (s)	3.5	3.5	3.5	3.5	3.5	3.5
All-Red Time (s)	2.0	2.0	2.0	2.0	2.0	2.0
Lost Time Adjust (s)	0.0	0.0	0.0	0.0	0.0	0.0
Total Lost Time (s)	5.5	5.5	5.5	5.5	5.5	5.5
Lead/Lag			Lead		Lag	Lag
Lead-Lag Optimize?			Yes		Yes	Yes
Recall Mode	None	None	None	C-Min	C-Min	C-Min
v/c Ratio	0.83	0.61	0.45	0.31	0.01	0.52
Control Delay (s/veh)	44.1	9.5	9.5	6.5	14.7	16.8
Queue Delay	0.0	0.0	0.0	0.0	0.0	0.0
Total Delay (s/veh)	44.1	9.5	9.5	6.5	14.7	16.8
Queue Length 50th (ft)	145	0	34	97	1	240
Queue Length 95th (ft)	230	76	78	174	m7	404
Internal Link Dist (ft)	433			445		2820
Turn Bay Length (ft)	135		305		190	
Base Capacity (vph)	555	651	485	2609	384	2112
Starvation Cap Reductn	0	0	0	0	0	0
Spillback Cap Reductn	0	0	0	0	0	0
Storage Cap Reductn	0	0	0	0	0	0
Reduced v/c Ratio	0.59	0.49	0.36	0.31	0.01	0.52

Intersection Summary

Cycle Length: 120
 Actuated Cycle Length: 120
 Offset: 0 (0%), Referenced to phase 2:SBTU and 6:NBTL, Start of Green
 Natural Cycle: 80
 Control Type: Actuated-Coordinated
 m Volume for 95th percentile queue is metered by upstream signal.

Splits and Phases: 1: Ernest W. Barrett Pkwy & Macedonia Rd



HCM 6th Signalized Intersection Summary
1: Ernest W. Barrett Pkwy & Macedonia Rd

3a. Future Build 2027 AM
11/18/2025

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	68	521	157	741	3	972	27
Future Volume (veh/h)	68	521	157	741	3	972	27
Initial Q (Qb), veh	0	0	0	0		0	0
Ped-Bike Adj(A_pbT)	1.00	1.00	1.00				1.00
Parking Bus, Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	653	173	814		1068	30
Peak Hour Factor	0.91	0.91	0.91	0.91		0.91	0.91
Percent Heavy Veh, %	2	2	2	2		2	2
Cap, veh/h	409	729	362	2411		2033	57
Arrive On Green	0.00	0.23	0.06	0.68		0.58	0.58
Sat Flow, veh/h	1781	3170	1781	3647		3624	99
Grp Volume(v), veh/h	0	653	173	814		538	560
Grp Sat Flow(s),veh/h/ln	1781	1585	1781	1777		1777	1853
Q Serve(g_s), s	0.0	24.0	4.5	11.5		22.1	22.1
Cycle Q Clear(g_c), s	0.0	24.0	4.5	11.5		22.1	22.1
Prop In Lane	1.00	1.00	1.00				0.05
Lane Grp Cap(c), veh/h	409	729	362	2411		1023	1067
V/C Ratio(X)	0.00	0.90	0.48	0.34		0.53	0.53
Avail Cap(c_a), veh/h	497	885	551	2411		1023	1067
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Upstream Filter(I)	0.00	1.00	1.00	1.00		1.00	1.00
Uniform Delay (d), s/veh	0.0	44.8	11.8	8.0		15.5	15.5
Incr Delay (d2), s/veh	0.0	10.3	1.0	0.4		1.9	1.9
Initial Q Delay(d3), s/veh	0.0	0.0	0.0	0.0		0.0	0.0
%ile BackOfQ(50%),veh/ln	0.0	10.3	1.6	3.9		8.7	9.0
Unsig. Movement Delay, s/veh							
LnGrp Delay(d), s/veh	0.0	55.1	12.8	8.4		17.4	17.3
LnGrp LOS		E	B	A		B	B
Approach Vol, veh/h	653			987		1098	
Approach Delay, s/veh	55.1			9.2		17.4	
Approach LOS	E			A		B	
Timer - Assigned Phs	1	2				6	8
Phs Duration (G+Y+Rc), s	12.3	74.6				86.9	33.1
Change Period (Y+Rc), s	5.5	5.5				5.5	5.5
Max Green Setting (Gmax), s	19.5	50.5				75.5	33.5
Max Q Clear Time (g_c+I1), s	6.5	24.1				13.5	26.0
Green Ext Time (p_c), s	0.3	10.3				9.0	1.6
Intersection Summary							
HCM 6th Ctrl Delay, s/veh			23.4				
HCM 6th LOS			C				
Notes							
User approved volume balancing among the lanes for turning movement.							
User approved ignoring U-Turning movement.							

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

3a. Future Build 2027 AM
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Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR		
Lane Configurations									
Traffic Volume (veh/h)	68	521	157	741	3	972	27		
Future Volume (veh/h)	68	521	157	741	3	972	27		
Number	3	18	1	6		2	12		
Initial Q, veh	0	0	0	0		0	0		
Ped-Bike Adj (A_pbT)	1.00	1.00	1.00				1.00		
Parking Bus Adj	1.00	1.00	1.00	1.00		1.00	1.00		
Work Zone On Approach	No			No		No			
Lanes Open During Work Zone									
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870		
Adj Flow Rate, veh/h	0	653	173	814		1068	30		
Peak Hour Factor	0.91	0.91	0.91	0.91		0.91	0.91		
Percent Heavy Veh, %	2	2	2	2		2	2		
Opposing Right Turn Influence	Yes		Yes						
Cap, veh/h	409	729	362	2411		2033	57		
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00		
Prop Arrive On Green	0.00	0.23	0.06	0.68		0.58	0.58		
Unsig. Movement Delay									
Ln Grp Delay, s/veh	0.0	55.1	12.8	8.4		17.4	17.3		
Ln Grp LOS		E	B	A		B	B		
Approach Vol, veh/h	653			987		1098			
Approach Delay, s/veh	55.1			9.2		17.4			
Approach LOS	E			A		B			
Timer:		1	2	3	4	5	6	7	8
Assigned Phs		1	2	8			6		
Case No		1.2	8.0	9.0			4.0		
Phs Duration (G+Y+Rc), s		12.3	74.6	33.1			86.9		
Change Period (Y+Rc), s		5.5	5.5	5.5			5.5		
Max Green (Gmax), s		19.5	50.5	33.5			75.5		
Max Allow Headway (MAH), s		3.5	5.8	3.8			5.8		
Max Q Clear (g_c+I1), s		6.5	24.1	26.0			13.5		
Green Ext Time (g_e), s		0.3	10.3	1.6			9.0		
Prob of Phs Call (p_c)		1.00	1.00	1.00			1.00		
Prob of Max Out (p_x)		0.00	0.24	0.23			0.00		
Left-Turn Movement Data									
Assigned Mvmt		1	5	3					
Mvmt Sat Flow, veh/h		1781	0	1781					
Through Movement Data									
Assigned Mvmt			2	8			6		
Mvmt Sat Flow, veh/h			3624	0			3647		
Right-Turn Movement Data									
Assigned Mvmt			12	18			16		
Mvmt Sat Flow, veh/h			99	3170			0		
Left Lane Group Data									
Assigned Mvmt		1	5	3	0	0	0	0	0
Lane Assignment		L (Pr/Pm)		L					

HCM 6th Signalized Intersection Capacity Analysis
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Lanes in Grp	1	0	1	0	0	0	0	0
Grp Vol (v), veh/h	173	0	0	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	1781	0	1781	0	0	0	0	0
Q Serve Time (g_s), s	4.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	4.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Sat Flow (s_l), veh/h/ln	514	0	1781	0	0	0	0	0
Shared LT Sat Flow (s_sh), veh/h/ln	0	0	0	0	0	0	0	0
Perm LT Eff Green (g_p), s	71.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Serve Time (g_u), s	47.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Q Serve Time (g_ps), s	12.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Time to First Blk (g_f), s	0.0	69.1	0.0	0.0	0.0	0.0	0.0	0.0
Serve Time pre Blk (g_fs), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop LT Inside Lane (P_L)	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	362	0	409	0	0	0	0	0
V/C Ratio (X)	0.48	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	551	0	497	0	0	0	0	0
Upstream Filter (I)	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	11.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	12.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
%ile Back of Q (50%), veh/ln	1.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Middle Lane Group Data								
Assigned Mvmt	0	2	8	0	0	6	0	0
Lane Assignment	T			T				
Lanes in Grp	0	1	0	0	0	2	0	0
Grp Vol (v), veh/h	0	538	0	0	0	814	0	0
Grp Sat Flow (s), veh/h/ln	0	1777	0	0	0	1777	0	0
Q Serve Time (g_s), s	0.0	22.1	0.0	0.0	0.0	11.5	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	22.1	0.0	0.0	0.0	11.5	0.0	0.0
Lane Grp Cap (c), veh/h	0	1023	0	0	0	2411	0	0
V/C Ratio (X)	0.00	0.53	0.00	0.00	0.00	0.34	0.00	0.00
Avail Cap (c_a), veh/h	0	1023	0	0	0	2411	0	0
Upstream Filter (I)	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	15.5	0.0	0.0	0.0	8.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.9	0.0	0.0	0.0	0.4	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	17.4	0.0	0.0	0.0	8.4	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	8.1	0.0	0.0	0.0	3.8	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.0	0.0	0.0	0.1	0.0	0.0

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3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	8.7	0.0	0.0	0.0	3.9	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.08	0.00	0.00	0.00	0.20	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Right Lane Group Data

Assigned Mvmt	0	12	18	0	0	16	0	0
Lane Assignment		T+R	R					
Lanes in Grp	0	1	2	0	0	0	0	0
Grp Vol (v), veh/h	0	560	653	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	0	1853	1585	0	0	0	0	0
Q Serve Time (g_s), s	0.0	22.1	24.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	22.1	24.0	0.0	0.0	0.0	0.0	0.0
Prot RT Sat Flow (s_R), veh/h/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prot RT Eff Green (g_R), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop RT Outside Lane (P_R)	0.00	0.05	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	0	1067	729	0	0	0	0	0
V/C Ratio (X)	0.00	0.53	0.90	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	0	1067	885	0	0	0	0	0
Upstream Filter (I)	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	15.5	44.8	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.9	10.3	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	17.3	55.1	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	8.5	9.3	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	1.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	9.0	10.3	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.08	0.57	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Intersection Summary

HCM 6th Ctrl Delay, s/veh	23.4
HCM 6th LOS	C

Notes

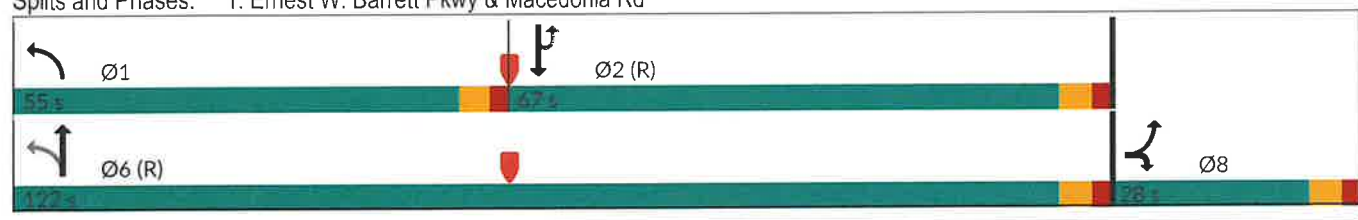
User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

						
Lane Group	EBL	EBR	NBL	NBT	SBU	SBT
Lane Configurations						
Traffic Volume (vph)	23	259	512	1245	8	961
Future Volume (vph)	23	259	512	1245	8	961
Lane Group Flow (vph)	153	150	551	1339	9	1096
Turn Type	Prot	Prot	pm+pt	NA	Perm	NA
Protected Phases	8	8	1	6		2
Permitted Phases			6		2	
Detector Phase	8	8	1	6	2	2
Switch Phase						
Minimum Initial (s)	6.0	6.0	5.0	15.0	15.0	15.0
Minimum Split (s)	23.5	23.5	15.0	23.5	40.5	40.5
Total Split (s)	28.0	28.0	55.0	122.0	67.0	67.0
Total Split (%)	18.7%	18.7%	36.7%	81.3%	44.7%	44.7%
Yellow Time (s)	3.5	3.5	3.5	3.5	3.5	3.5
All-Red Time (s)	2.0	2.0	2.0	2.0	2.0	2.0
Lost Time Adjust (s)	0.0	0.0	0.0	0.0	0.0	0.0
Total Lost Time (s)	5.5	5.5	5.5	5.5	5.5	5.5
Lead/Lag			Lead		Lag	Lag
Lead-Lag Optimize?			Yes		Yes	Yes
Recall Mode	None	None	None	C-Min	C-Min	C-Min
v/c Ratio	0.70	0.64	0.79	0.44	0.05	0.59
Control Delay (s/veh)	32.5	22.0	31.9	2.9	24.1	28.0
Queue Delay	0.0	0.0	0.0	0.0	0.0	0.0
Total Delay (s/veh)	32.5	22.0	31.9	2.9	24.1	28.0
Queue Length 50th (ft)	24	0	294	102	4	372
Queue Length 95th (ft)	97	73	434	190	m18	564
Internal Link Dist (ft)	433			445		2820
Turn Bay Length (ft)	135		305		190	
Base Capacity (vph)	351	353	752	3059	199	1848
Starvation Cap Reductn	0	0	0	0	0	0
Spillback Cap Reductn	0	0	0	0	0	0
Storage Cap Reductn	0	0	0	0	0	0
Reduced v/c Ratio	0.44	0.42	0.73	0.44	0.05	0.59

Intersection Summary

Cycle Length: 150
 Actuated Cycle Length: 150
 Offset: 0 (0%), Referenced to phase 2:SBTU and 6:NBTL, Start of Green
 Natural Cycle: 90
 Control Type: Actuated-Coordinated
 m Volume for 95th percentile queue is metered by upstream signal.

Splits and Phases: 1: Ernest W. Barrett Pkwy & Macedonia Rd



HCM 6th Signalized Intersection Summary
1: Ernest W. Barrett Pkwy & Macedonia Rd

3b. Future Build 2027 PM
11/18/2025

							
Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	23	259	512	1245	8	961	59
Future Volume (veh/h)	23	259	512	1245	8	961	59
Initial Q (Qb), veh	0	0	0	0		0	0
Ped-Bike Adj(A_pbT)	1.00	1.00	1.00				1.00
Parking Bus, Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	305	551	1339		1033	63
Peak Hour Factor	0.93	0.93	0.93	0.93		0.93	0.93
Percent Heavy Veh, %	2	2	2	2		2	2
Cap, veh/h	199	354	580	2896		2057	125
Arrive On Green	0.00	0.11	0.17	0.81		0.60	0.60
Sat Flow, veh/h	1781	3170	1781	3647		3496	207
Grp Volume(v), veh/h	0	305	551	1339		539	557
Grp Sat Flow(s),veh/h/ln	1781	1585	1781	1777		1777	1833
Q Serve(g_s), s	0.0	14.2	22.5	16.8		25.9	25.9
Cycle Q Clear(g_c), s	0.0	14.2	22.5	16.8		25.9	25.9
Prop In Lane	1.00	1.00	1.00				0.11
Lane Grp Cap(c), veh/h	199	354	580	2896		1074	1108
V/C Ratio(X)	0.00	0.86	0.95	0.46		0.50	0.50
Avail Cap(c_a), veh/h	267	476	858	2896		1074	1108
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Upstream Filter(I)	0.00	1.00	1.00	1.00		1.00	1.00
Uniform Delay (d), s/veh	0.0	65.5	26.6	4.1		16.8	16.8
Incr Delay (d2), s/veh	0.0	11.7	15.4	0.5		1.7	1.6
Initial Q Delay(d3), s/veh	0.0	0.0	0.0	0.0		0.0	0.0
%ile BackOfQ(50%),veh/ln	0.0	6.3	22.1	4.7		10.5	10.8
Unsig. Movement Delay, s/veh							
LnGrp Delay(d), s/veh	0.0	77.2	42.0	4.7		18.5	18.5
LnGrp LOS		E	D	A		B	B
Approach Vol, veh/h	305			1890		1096	
Approach Delay, s/veh	77.2			15.5		18.5	
Approach LOS	E			B		B	
Timer - Assigned Phs	1	2				6	8
Phs Duration (G+Y+Rc), s	31.6	96.2				127.7	22.3
Change Period (Y+Rc), s	5.5	5.5				5.5	5.5
Max Green Setting (Gmax), s	49.5	61.5				116.5	22.5
Max Q Clear Time (g_c+l1), s	24.5	27.9				18.8	16.2
Green Ext Time (p_c), s	1.5	11.3				21.4	0.6
Intersection Summary							
HCM 6th Ctrl Delay, s/veh			22.2				
HCM 6th LOS			C				
Notes							
User approved volume balancing among the lanes for turning movement.							
User approved ignoring U-Turning movement.							

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

3b. Future Build 2027 PM
11/18/2025

									
Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR		
Lane Configurations									
Traffic Volume (veh/h)	23	259	512	1245	8	961	59		
Future Volume (veh/h)	23	259	512	1245	8	961	59		
Number	3	18	1	6		2	12		
Initial Q, veh	0	0	0	0		0	0		
Ped-Bike Adj (A_pbT)	1.00	1.00	1.00				1.00		
Parking Bus Adj	1.00	1.00	1.00	1.00		1.00	1.00		
Work Zone On Approach	No			No		No			
Lanes Open During Work Zone									
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870		
Adj Flow Rate, veh/h	0	305	551	1339		1033	63		
Peak Hour Factor	0.93	0.93	0.93	0.93		0.93	0.93		
Percent Heavy Veh, %	2	2	2	2		2	2		
Opposing Right Turn Influence	Yes		Yes						
Cap, veh/h	199	354	580	2896		2057	125		
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00		
Prop Arrive On Green	0.00	0.11	0.17	0.81		0.60	0.60		
Unsig. Movement Delay									
Ln Grp Delay, s/veh	0.0	77.2	42.0	4.7		18.5	18.5		
Ln Grp LOS		E	D	A		B	B		
Approach Vol, veh/h	305			1890		1096			
Approach Delay, s/veh	77.2			15.5		18.5			
Approach LOS	E			B		B			
Timer:		1	2	3	4	5	6	7	8
Assigned Phs		1	2	8			6		
Case No		1.2	8.0	9.0			4.0		
Phs Duration (G+Y+Rc), s		31.6	96.2	22.3			127.7		
Change Period (Y+Rc), s		5.5	5.5	5.5			5.5		
Max Green (Gmax), s		49.5	61.5	22.5			116.5		
Max Allow Headway (MAH), s		3.5	5.8	3.8			5.8		
Max Q Clear (g_c+I1), s		24.5	27.9	16.2			18.8		
Green Ext Time (g_e), s		1.5	11.3	0.6			21.4		
Prob of Phs Call (p_c)		1.00	1.00	1.00			1.00		
Prob of Max Out (p_x)		0.00	0.14	0.17			0.01		
Left-Turn Movement Data									
Assigned Mvmt		1	5	3					
Mvmt Sat Flow, veh/h		1781	0	1781					
Through Movement Data									
Assigned Mvmt			2	8			6		
Mvmt Sat Flow, veh/h			3496	0			3647		
Right-Turn Movement Data									
Assigned Mvmt			12	18			16		
Mvmt Sat Flow, veh/h			207	3170			0		
Left Lane Group Data									
Assigned Mvmt		1	5	3	0	0	0	0	0
Lane Assignment		L (Pr/Pm)			L				

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

3b. Future Build 2027 PM
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Lanes in Grp	1	0	1	0	0	0	0	0
Grp Vol (v), veh/h	551	0	0	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	1781	0	1781	0	0	0	0	0
Q Serve Time (g_s), s	22.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	22.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Sat Flow (s_l), veh/h/ln	515	0	1781	0	0	0	0	0
Shared LT Sat Flow (s_sh), veh/h/ln	0	0	0	0	0	0	0	0
Perm LT Eff Green (g_p), s	92.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Serve Time (g_u), s	64.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Q Serve Time (g_ps), s	64.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Time to First Blk (g_f), s	0.0	90.7	0.0	0.0	0.0	0.0	0.0	0.0
Serve Time pre Blk (g_fs), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop LT Inside Lane (P_L)	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	580	0	199	0	0	0	0	0
V/C Ratio (X)	0.95	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	858	0	267	0	0	0	0	0
Upstream Filter (I)	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	26.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	15.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	42.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	19.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	2.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
%ile Back of Q (50%), veh/ln	22.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	1.84	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Middle Lane Group Data

Assigned Mvmt	0	2	8	0	0	6	0	0
Lane Assignment		T				T		
Lanes in Grp	0	1	0	0	0	2	0	0
Grp Vol (v), veh/h	0	539	0	0	0	1339	0	0
Grp Sat Flow (s), veh/h/ln	0	1777	0	0	0	1777	0	0
Q Serve Time (g_s), s	0.0	25.9	0.0	0.0	0.0	16.8	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	25.9	0.0	0.0	0.0	16.8	0.0	0.0
Lane Grp Cap (c), veh/h	0	1074	0	0	0	2896	0	0
V/C Ratio (X)	0.00	0.50	0.00	0.00	0.00	0.46	0.00	0.00
Avail Cap (c_a), veh/h	0	1074	0	0	0	2896	0	0
Upstream Filter (I)	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	16.8	0.0	0.0	0.0	4.1	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.7	0.0	0.0	0.0	0.5	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	18.5	0.0	0.0	0.0	4.7	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	10.0	0.0	0.0	0.0	4.4	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.0	0.0	0.0	0.2	0.0	0.0

HCM 6th Signalized Intersection Capacity Analysis
1: Ernest W. Barrett Pkwy & Macedonia Rd

3b. Future Build 2027 PM
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3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	10.5	0.0	0.0	0.0	4.7	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.09	0.00	0.00	0.00	0.24	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Right Lane Group Data								
Assigned Mvmt	0	12	18	0	0	16	0	0
Lane Assignment		T+R	R					
Lanes in Grp	0	1	2	0	0	0	0	0
Grp Vol (v), veh/h	0	557	305	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	0	1833	1585	0	0	0	0	0
Q Serve Time (g_s), s	0.0	25.9	14.2	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	25.9	14.2	0.0	0.0	0.0	0.0	0.0
Prot RT Sat Flow (s_R), veh/h/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prot RT Eff Green (g_R), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop RT Outside Lane (P_R)	0.00	0.11	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	0	1108	354	0	0	0	0	0
V/C Ratio (X)	0.00	0.50	0.86	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	0	1108	476	0	0	0	0	0
Upstream Filter (I)	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	16.8	65.5	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.6	11.7	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	18.5	77.2	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	10.3	5.7	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.6	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	10.8	6.3	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.10	0.35	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Intersection Summary								
HCM 6th Ctrl Delay, s/veh	22.2							
HCM 6th LOS	C							

Notes								
User approved volume balancing among the lanes for turning movement.								
User approved ignoring U-Turning movement.								

TRAFFIC VOLUME WORKSHEETS

25-196 Res Dev at Old Horseshoe Bend Rd and Ernest Barrett Pkwy - Cobb County, GA
Traffic Volumes

A&R Engineering
 November 2025

1. Ernest W.B.Pkwy @ Macedonia

A.M. Peak Hour

Condition	Ernest W. Barrett Parkway						Ernest W. Barrett Parkway						Macedonia Road					
	Northbound						Southbound						Eastbound					
	U	L	T	R	Tot		U	L	T	R	Tot		U	L	T	R	Tot	
Existing 2025 Traffic Counts:	0	154	722	0	876		0	0	940	24	964		0	66	0	511	577	
Growth Factor (%):	1	1	1	1	1		1	1	1	1	1		1	1	1	1	1	
No-Build 2027 Volumes:	0	157	736	0	893		0	0	959	24	983		0	67	0	521	588	
Total New Trips:	0	0	5	0	5		3	0	13	3	19		0	1	0	0	1	
Future 2027 Traffic Volumes:	0	157	741	0	898		3	0	972	27	1002		0	68	0	521	589	

P.M. Peak Hour

Condition	Ernest W. Barrett Parkway						Ernest W. Barrett Parkway						Macedonia Road					
	Northbound						Southbound						Eastbound					
	U	L	T	R	Tot		U	L	T	R	Tot		U	L	T	R	Tot	
Existing 2025 Traffic Counts:	0	502	1205	0	1707		0	0	933	56	989		0	20	0	254	274	
Growth Factor (%):	1	1	1	1	1		1	1	1	1	1		1	1	1	1	1	
No-Build 2027 Volumes:	0	512	1229	0	1741		0	0	952	57	1009		0	20	0	259	279	
Total New Trips:	0	0	16	0	16		8	0	9	2	19		0	3	0	0	3	
Future 2027 Traffic Volumes:	0	512	1245	0	1757		8	0	961	59	1028		0	23	0	259	282	

25-196 Res Dev at Old Horseshoe Bend Rd and Ernest Barrett Pkwy - Cobb County, GA
Traffic Volumes

A&R Engineering
November 2025

2.Ernest W.B.Pkwy @ Median Brk

A.M. Peak Hour

Condition	Ernest W. Barrett Parkway						Ernest W. Barrett Parkway						Westbound					
	Northbound						Southbound						Eastbound					
	U	L	T	R	Tot		U	L	T	R	Tot		U	L	T	R	Tot	
Existing 2025 Traffic Counts:	1	0	787	0	788		3	0	964	0	967		0	0	0	0	0	0
Growth Factor (%):	1	1	1	1	1		1	1	1	1	1		1	1	1	1	1	1
No-Build 2027 Volumes:	1	0	803	0	804		3	0	983	0	986		0	0	0	0	0	0
Total New Trips:	15	0	10	0	25		0	0	3	0	3		0	0	0	0	0	0
Future 2027 Traffic Volumes:	16	0	813	0	829		3	0	986	0	989		0	0	0	0	0	0

P.M. Peak Hour

Condition	Ernest W. Barrett Parkway						Ernest W. Barrett Parkway						Westbound					
	Northbound						Southbound						Eastbound					
	U	L	T	R	Tot		U	L	T	R	Tot		U	L	T	R	Tot	
Existing 2025 Traffic Counts:	10	0	1305	0	1315		1	0	951	0	952		0	0	0	0	0	0
Growth Factor (%):	1	1	1	1	1		1	1	1	1	1		1	1	1	1	1	1
No-Build 2027 Volumes:	10	0	1331	0	1341		1	0	970	0	971		0	0	0	0	0	0
Total New Trips:	10	0	7	0	17		0	0	8	0	8		0	0	0	0	0	0
Future 2027 Traffic Volumes:	20	0	1338	0	1358		1	0	978	0	979		0	0	0	0	0	0

25-196 Res Dev at Old Horseshoe Bend Rd and Ernest Barrett Pkwy - Cobb County, GA
Traffic Volumes

A&R Engineering
 November 2025

3. Ernest W.B. Pkwy @ Drwy (RIRO)

A.M. Peak Hour

Condition	Ernest W. Barrett Parkway						Ernest W. Barrett Parkway						Eastbound						Site Driveway (Right-In/ Right-Out)					
	Northbound						Southbound						Eastbound						Westbound					
	U	L	T	R	Tot		U	L	T	R	Tot		U	L	T	R	Tot		U	L	T	R	Tot	
Existing 2025 Traffic Counts:	0	0	788	0	788		0	0	965	0	965		0	0	0	0	0		0	0	0	0	0	
Growth Factor (%):	1	1	1	1	1		1	1	1	1	1		1	1	1	1	1		1	1	1	1	1	
No-Build 2027 Volumes:	0	0	804	0	804		0	0	984	0	984		0	0	0	0	0		0	0	0	0	0	
Total New Trips:	0	0	0	9	9		0	0	18	0	18		0	0	0	0	0		0	0	0	0	0	
Future 2027 Traffic Volumes:	0	0	804	9	813		0	0	1002	0	1002		0	0	0	0	0		0	0	0	0	0	

P.M. Peak Hour

Condition	Ernest W. Barrett Parkway						Ernest W. Barrett Parkway						Eastbound						Site Driveway (Right-In/ Right-Out)					
	Northbound						Southbound						Eastbound						Westbound					
	U	L	T	R	Tot		U	L	T	R	Tot		U	L	T	R	Tot		U	L	T	R	Tot	
Existing 2025 Traffic Counts:	0	0	1315	0	1315		0	0	961	0	961		0	0	0	0	0		0	0	0	0	0	
Growth Factor (%):	1	1	1	1	1		1	1	1	1	1		1	1	1	1	1		1	1	1	1	1	
No-Build 2027 Volumes:	0	0	1341	0	1341		0	0	980	0	980		0	0	0	0	0		0	0	0	0	0	
Total New Trips:	0	0	0	27	27		0	0	18	0	18		0	0	0	0	0		0	0	0	0	0	
Future 2027 Traffic Volumes:	0	0	1341	27	1368		0	0	998	0	998		0	0	0	0	0		0	0	0	0	0	